BackEnd

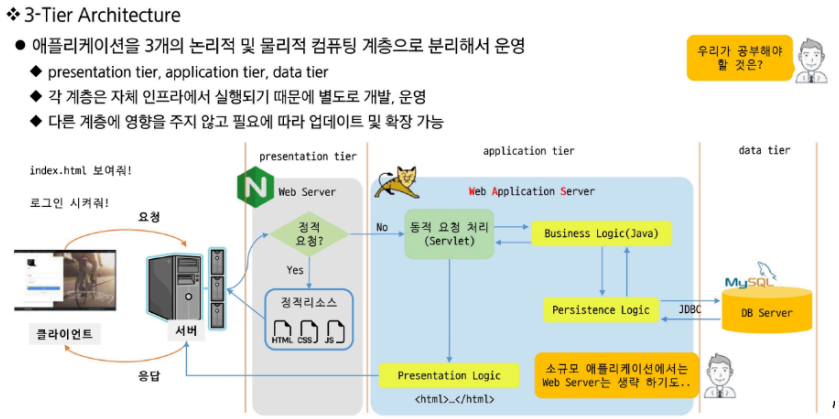
▼ 10/27 오전

1. Backend Programming
2. Maven
3. Servlet

Java-Web(Back) : Servlet1

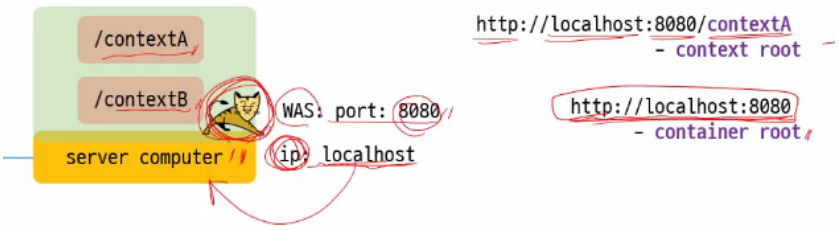
3-Tier Architecture

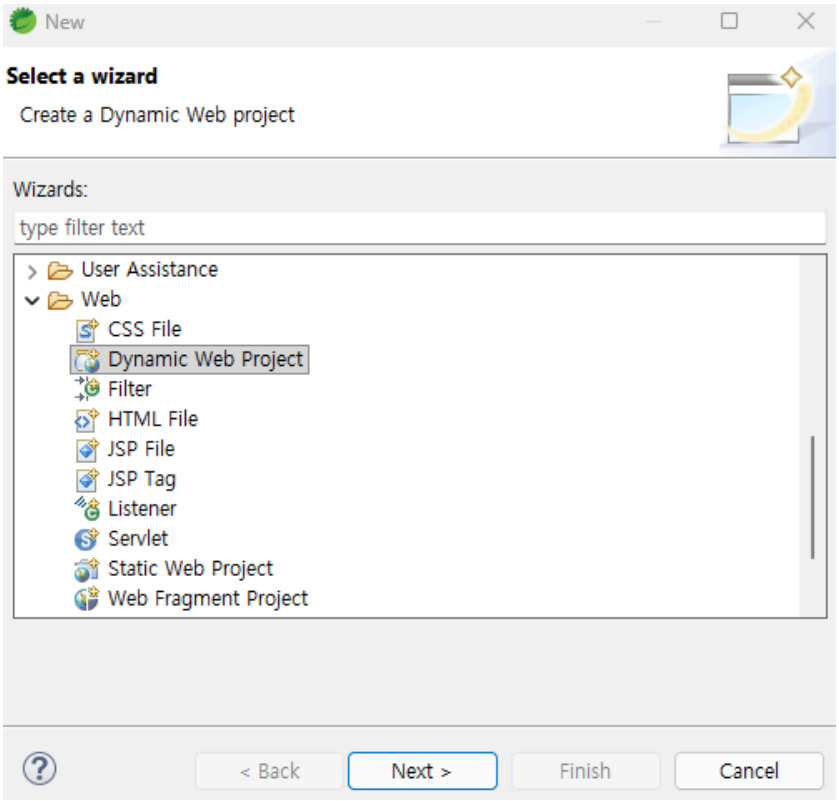
: presentation tier(Web Server), application tier(Web Application Server), data tier(DB)

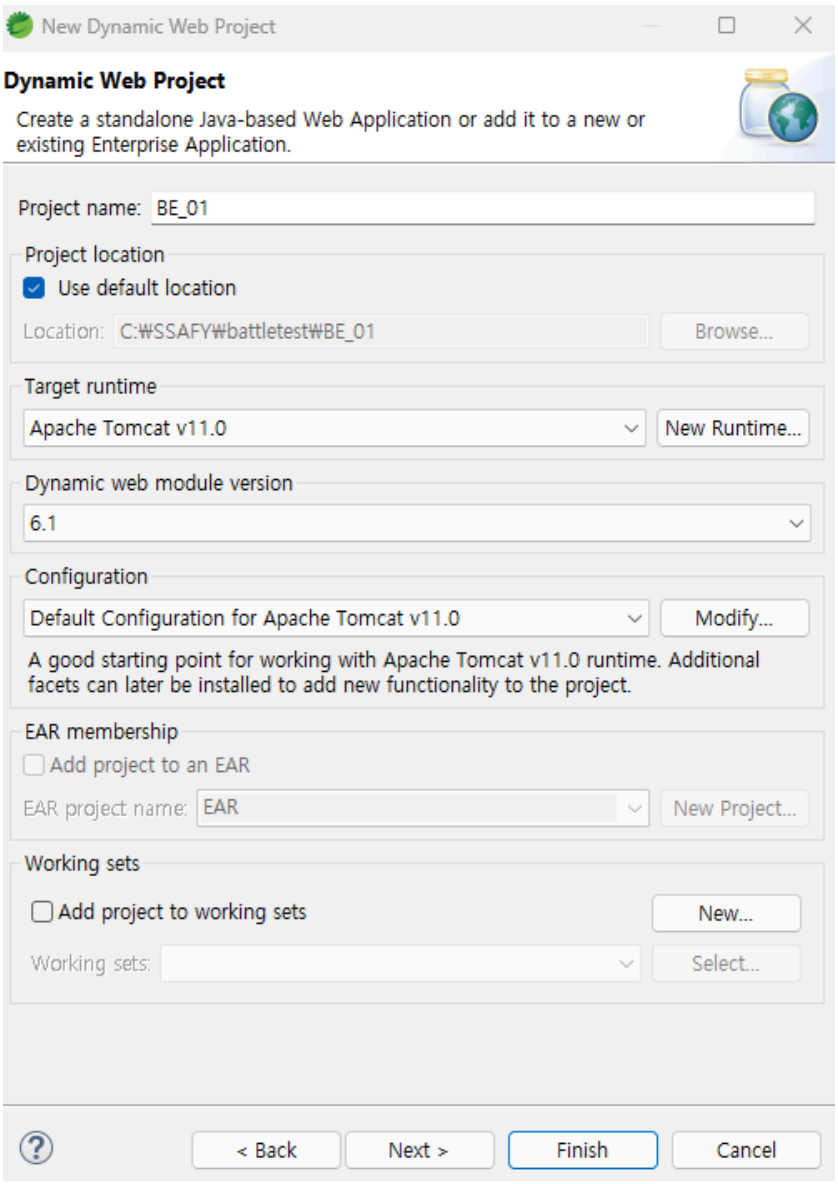


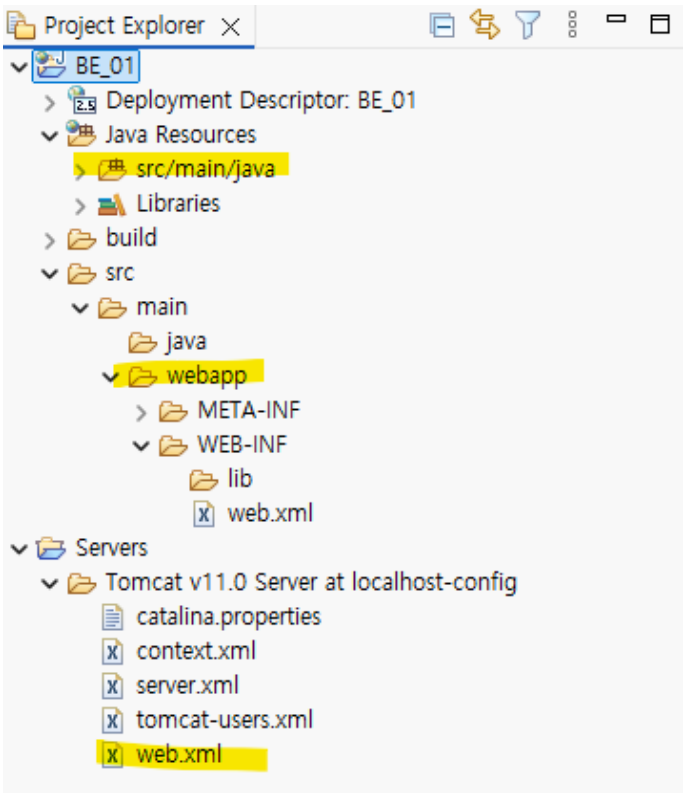
WAS : Web Application Server

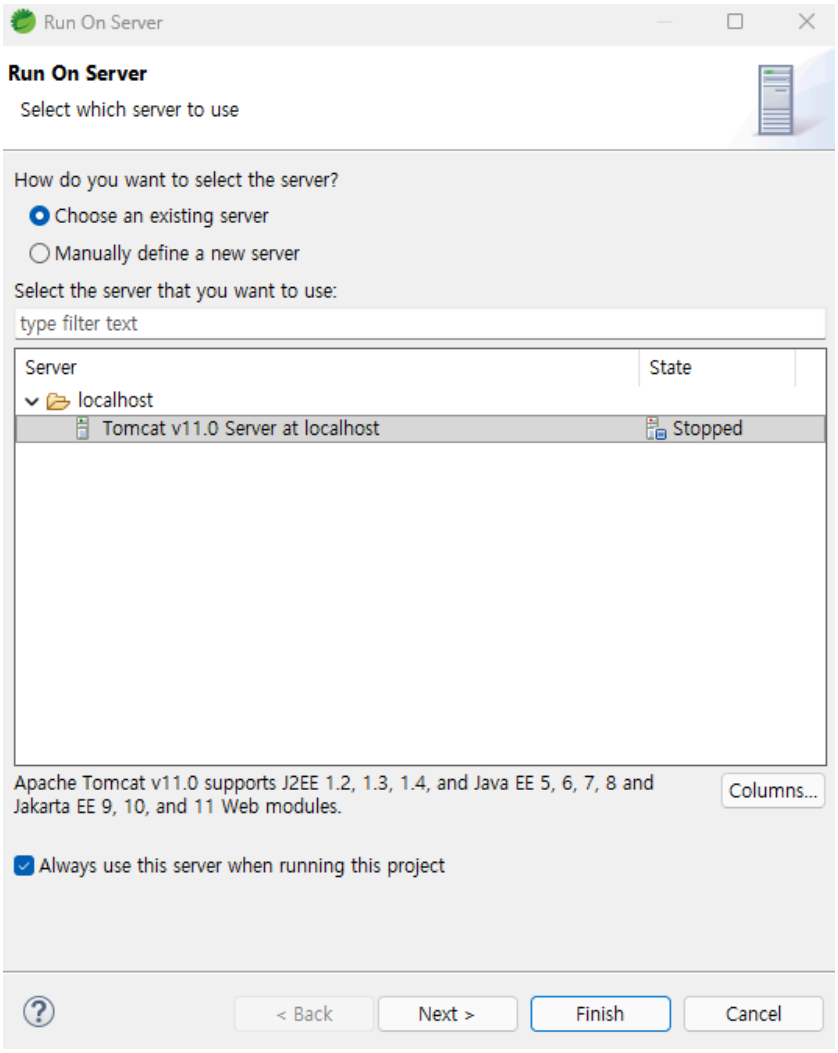
container / context 차이

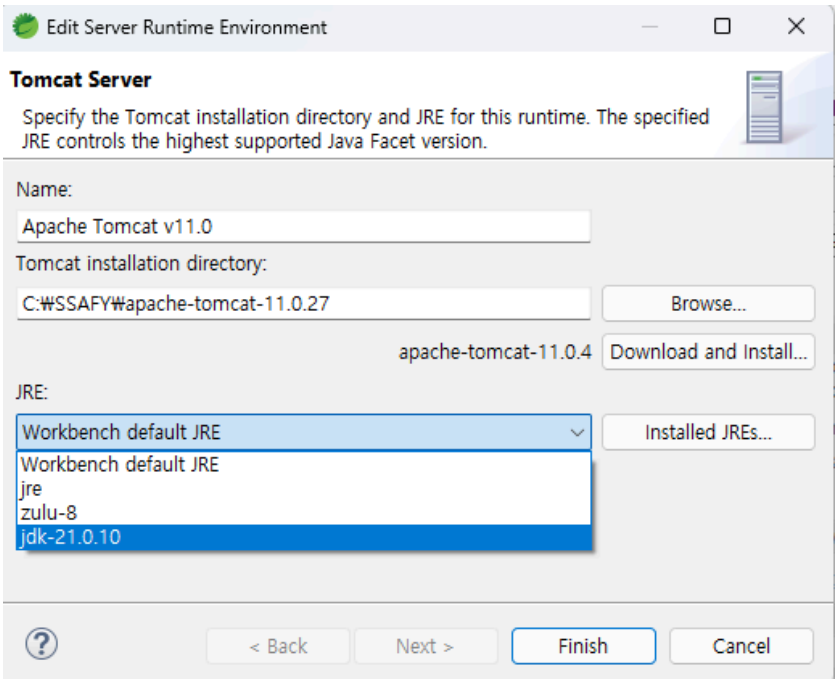

/contextA
/contextB
WAS: port: 8080
server computer
ip: localhost
http://localhost:8080/contextA
- context root
http://localhost:8080
- container root

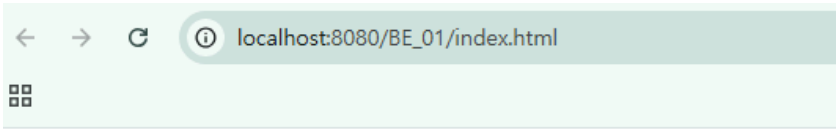

New
Select a wizard
Create a Dynamic Web project
Wizards:
type filter text
User Assistance
Web
CSS File
Dynamic Web Project
Filter
HTML File
JSP File
JSP Tag
Listener
Servlet
Static Web Project
Web Fragment Project
< Back
Next >
Finish
Cancel


New Dynamic Web Project
Dynamic Web Project
Create a standalone Java-based Web Application or add it to a new or existing Enterprise Application.
Project name: BE_01
Project location
Use default location
Location: C:₩SSAFY₩battletest₩BE_01
Browse...
Target runtime
Apache Tomcat v11.0
New Runtime...
Dynamic web module version
6.1
Configuration
Default Configuration for Apache Tomcat v11.0
Modify...
A good starting point for working with Apache Tomcat v11.0 runtime. Additional facets can later be installed to add new functionality to the project.
EAR membership
Add project to an EAR
EAR project name: EAR
New Project...
Working sets
Add project to working sets
New...
Working sets:
Select...
< Back
Next >
Finish
Cancel


Project Explorer
BE_01
Deployment Descriptor: BE_01
Java Resources
src/main/java
Libraries
build
src
main
java
webapp
META-INF
WEB-INF
lib
web.xml
Servers
Tomcat v11.0 Server at localhost-config
catalina.properties
context.xml
server.xml
tomcat-users.xml
web.xml


Run On Server
Run On Server
Select which server to use
How do you want to select the server?
Choose an existing server
Manually define a new server
Select the server that you want to use:
type filter text
Server
State
localhost
Tomcat v11.0 Server at localhost
Stopped
Apache Tomcat v11.0 supports J2EE 1.2, 1.3, 1.4, and Java EE 5, 6, 7, 8 and Jakarta EE 9, 10, and 11 Web modules.
Columns...
Always use this server when running this project
< Back
Next >
Finish
Cancel





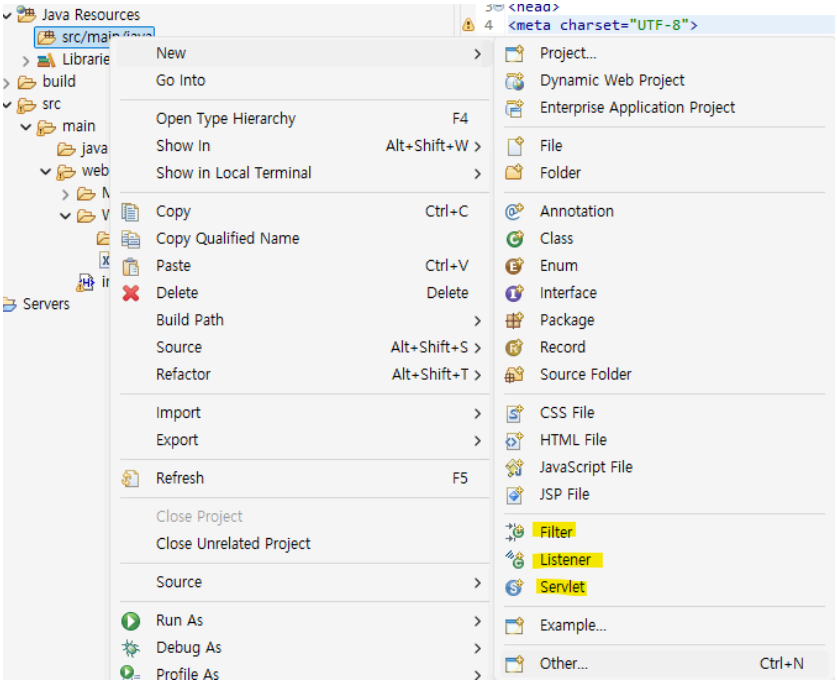
Hello Web

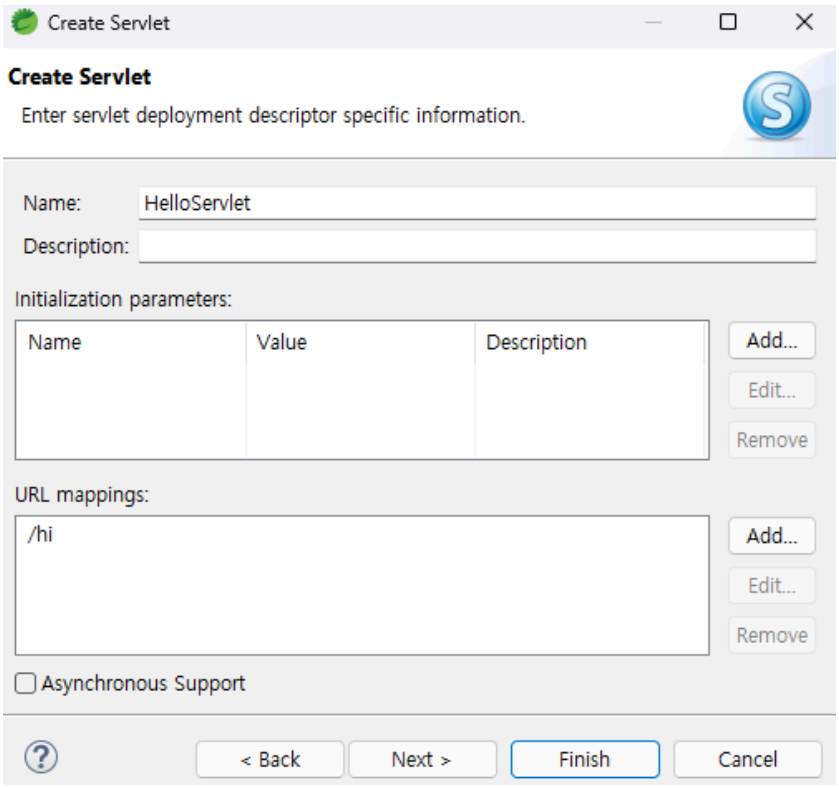
localhost 내 컴퓨터

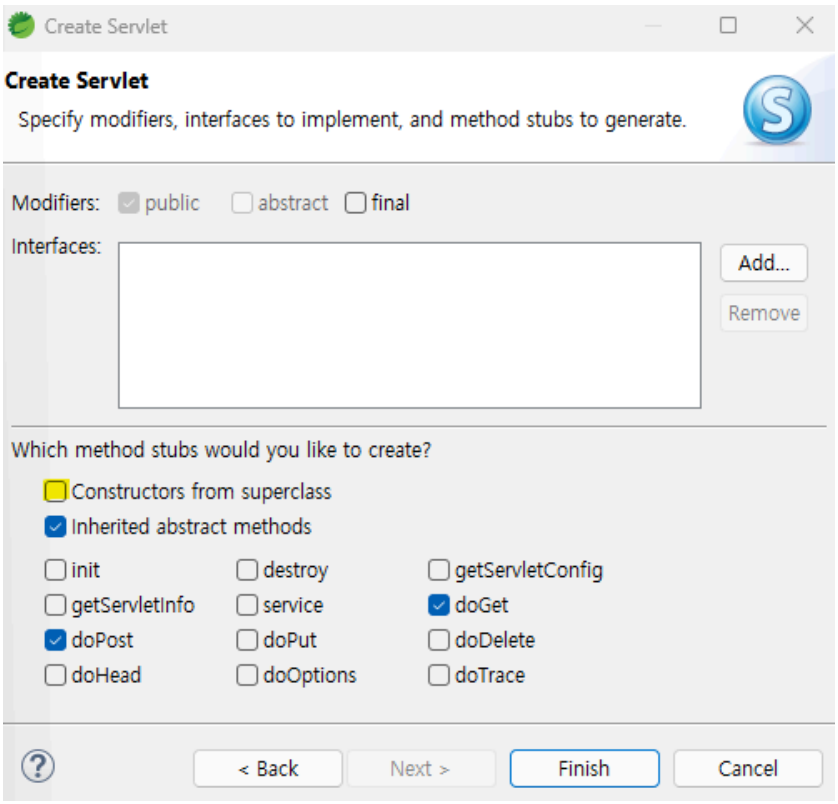
8080 was

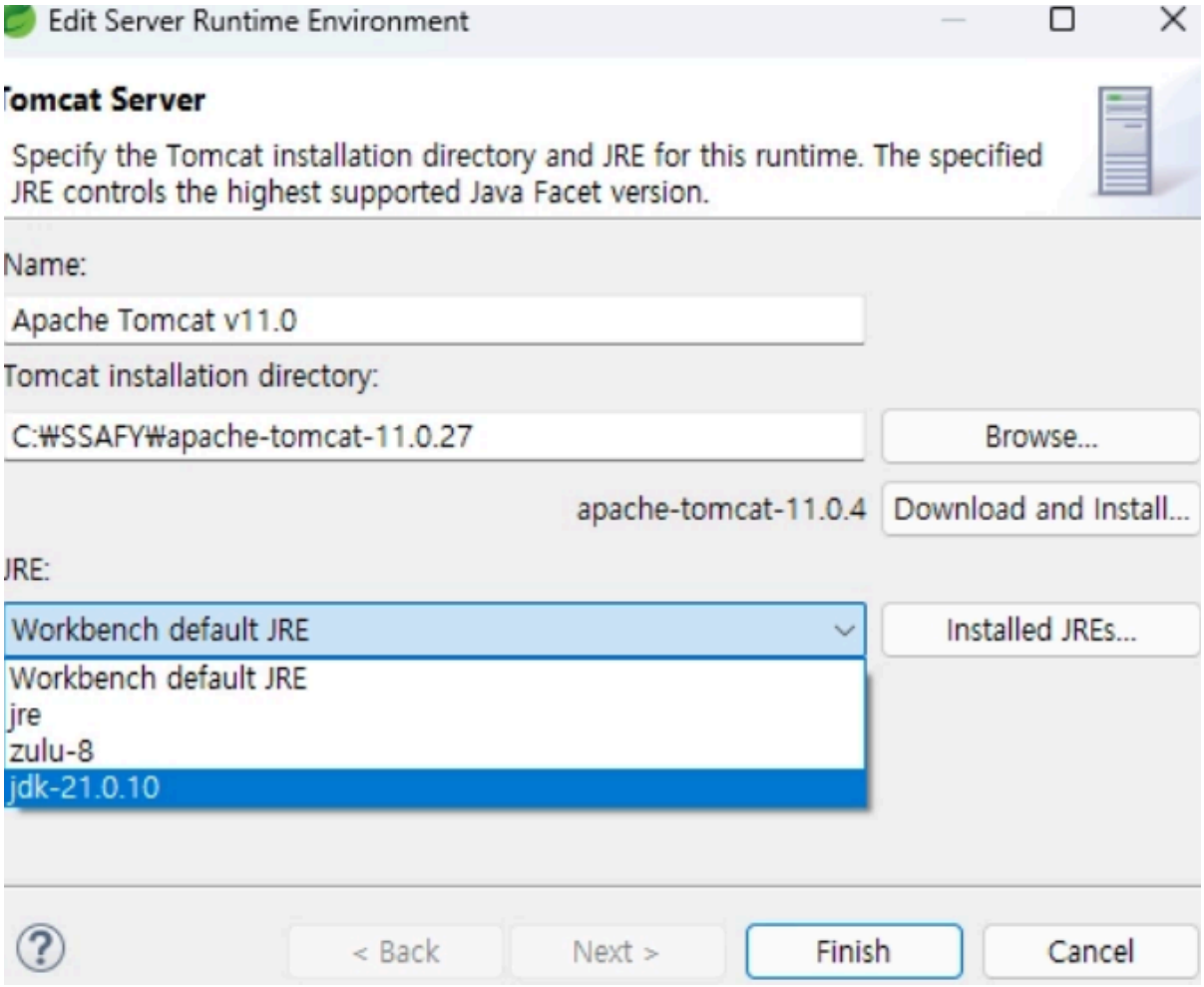
be_01 context

정적인 요소? 동적인 요소?


<head>
<meta charset="UTF-8">
Java Resources
src/main/java
Librarie
build
src
main
java
web
Servers
New
Go Into
Open Type Hierarchy F4
Show In Alt+Shift+W
Show in Local Terminal
Copy Ctrl+C
Copy Qualified Name
Paste Ctrl+V
Delete Delete
Build Path
Source Alt+Shift+S
Refactor Alt+Shift+T
Import
Export
Refresh F5
Close Project
Close Unrelated Project
Source
Run As
Debug As
Profile As
Project...
Dynamic Web Project
Enterprise Application Project
File
Folder
Annotation
Class
Enum
Interface
Package
Record
Source Folder
CSS File
HTML File
JavaScript File
JSP File
Filter
Listener
Servlet
Example...
Other... Ctrl+N


Create Servlet
Create Servlet
Enter servlet deployment descriptor specific information.
Name: HelloServlet
Description:
Initialization parameters:
Name
Value
Description
Add...
Edit...
Remove
URL mappings:
/hi
Add...
Edit...
Remove
Asynchronous Support
< Back
Next >
Finish
Cancel


Create Servlet
Create Servlet
Specify modifiers, interfaces to implement, and method stubs to generate.
Modifiers: public abstract final
Interfaces:
Add...
Remove
Which method stubs would you like to create?
Constructors from superclass
Inherited abstract methods
init
destroy
getServletConfig
getServletInfo
service
doGet
doPost
doPut
doDelete
doHead
doOptions
doTrace
< Back
Next >
Finish
Cancel

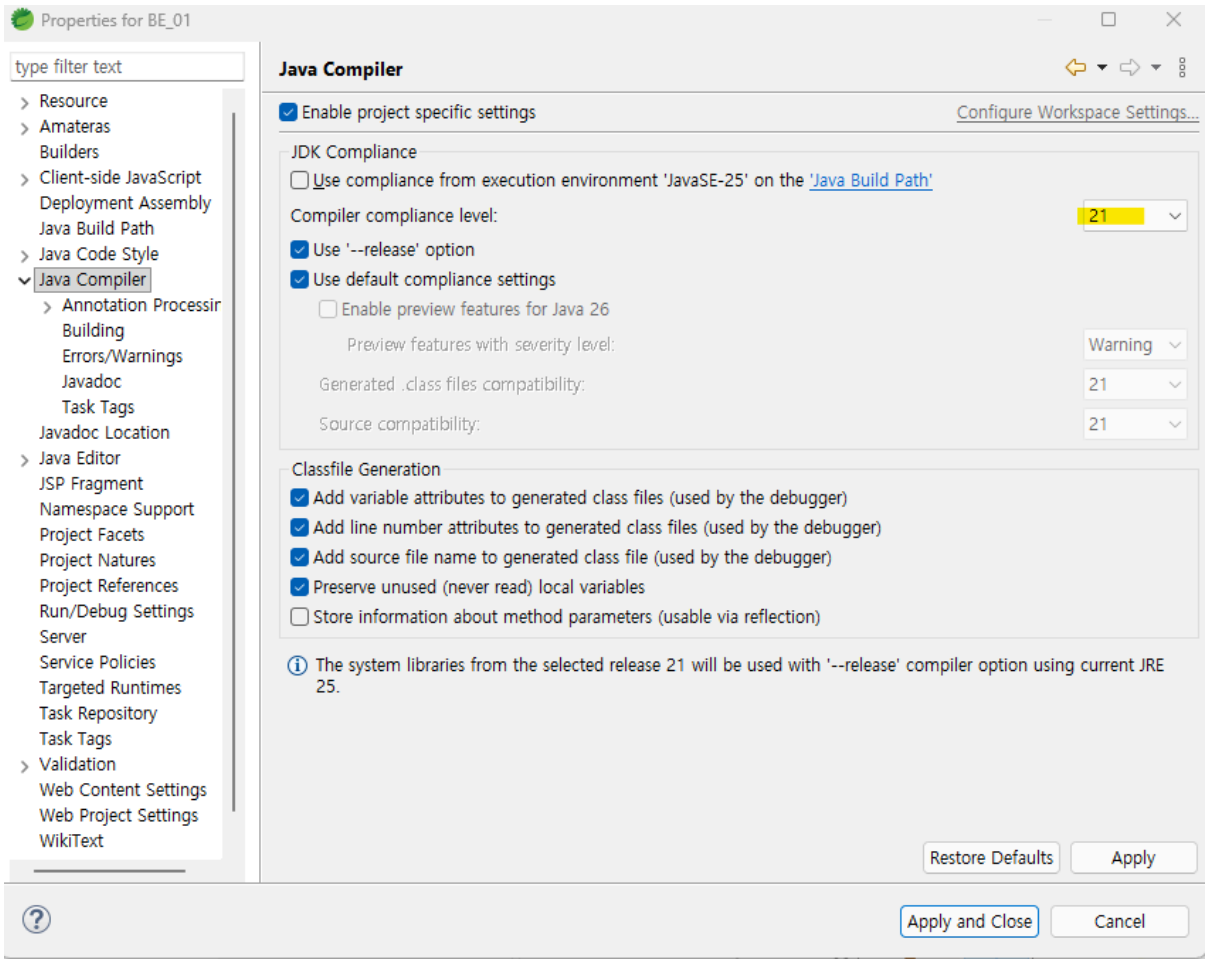


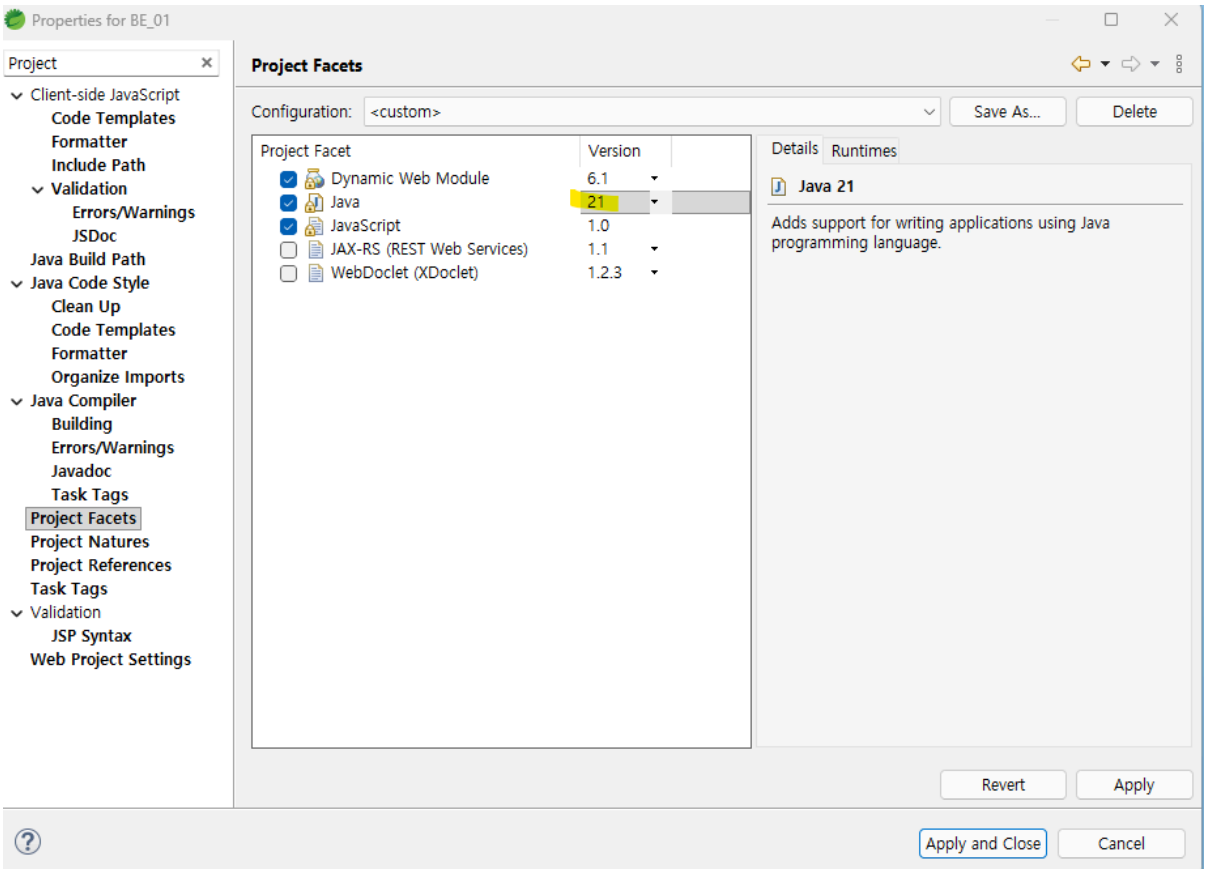
Preferences 메뉴에서 확인

이클립스 상단 메뉴에서 Window > Preferences를 클릭합니다.

좌측 메뉴에서 Server > Runtime Environments로 이동합니다.

목록에서 Apache Tomcat v11.0을 선택하고 우측의 Edit... 버튼을 클릭하여 JRE 버전을 확인하고 변경할 수 있습니다.


Properties for BE_01
type filter text
Resource
Amateras
Builders
Client-side JavaScript
Deployment Assembly
Java Build Path
Java Code Style
Java Compiler
Annotation Processir
Building
Errors/Warnings
Javadoc
Task Tags
Javadoc Location
Java Editor
JSP Fragment
Namespace Support
Project Facets
Project Natures
Project References
Run/Debug Settings
Server
Service Policies
Targeted Runtimes
Task Repository
Task Tags
Validation
Web Content Settings
Web Project Settings
WikiText
Java Compiler
Enable project specific settings
Configure Workspace Settings...
JDK Compliance
Use compliance from execution environment 'JavaSE-25' on the 'Java Build Path'
Compiler compliance level:
21
Use '--release' option
Use default compliance settings
Enable preview features for Java 26
Preview features with severity level:
Warning
Generated .class files compatibility:
21
Source compatibility:
21
Classfile Generation
Add variable attributes to generated class files (used by the debugger)
Add line number attributes to generated class files (used by the debugger)
Add source file name to generated class file (used by the debugger)
Preserve unused (never read) local variables
Store information about method parameters (usable via reflection)
The system libraries from the selected release 21 will be used with '--release' compiler option using current JRE 25.
Restore Defaults
Apply
Apply and Close
Cancel



에러 로그의 `UnsupportedClassVersionError`는 **프로젝트가 컴파일된 자바 버전과 톰캣 서버가 실행되는 자바 버전이 맞지 않아 발생하는 충돌**입니다.

작성한 코드는 더 높은 자바 버전(class file version 69.0)으로 컴파일되었는데, 앞서 톰캣 서버에 설정한 환경은 자바 21(최대 65.0까지만 인식)이기 때문에 클래스 파일을 읽어 들이지 못하는 상태입니다. 이클립스 프로젝트의 자바 버전을 서버 환경(Java 21)에 맞춰주어야 합니다.

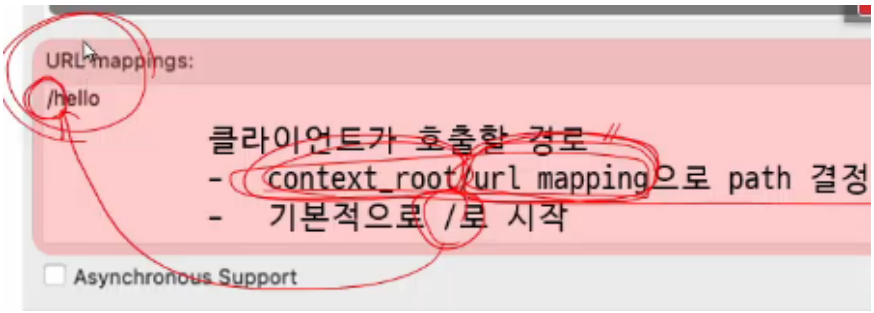
Java Compiler 버전 변경

1. 좌측 Project Explorer에서 `BE_01` 프로젝트를 우클릭하고 맨 아래의 **Properties**(속성)를 클릭합니다.
2. 창이 열리면 좌측 메뉴에서 **Java Compiler**를 선택합니다.
3. 우측 화면에서 'Compiler compliance level'을 **21**로 변경합니다. (만약 비활성화되어 있다면 바로 위의 'Use compliance from execution environment...' 체크를 해제하면 됩니다.)

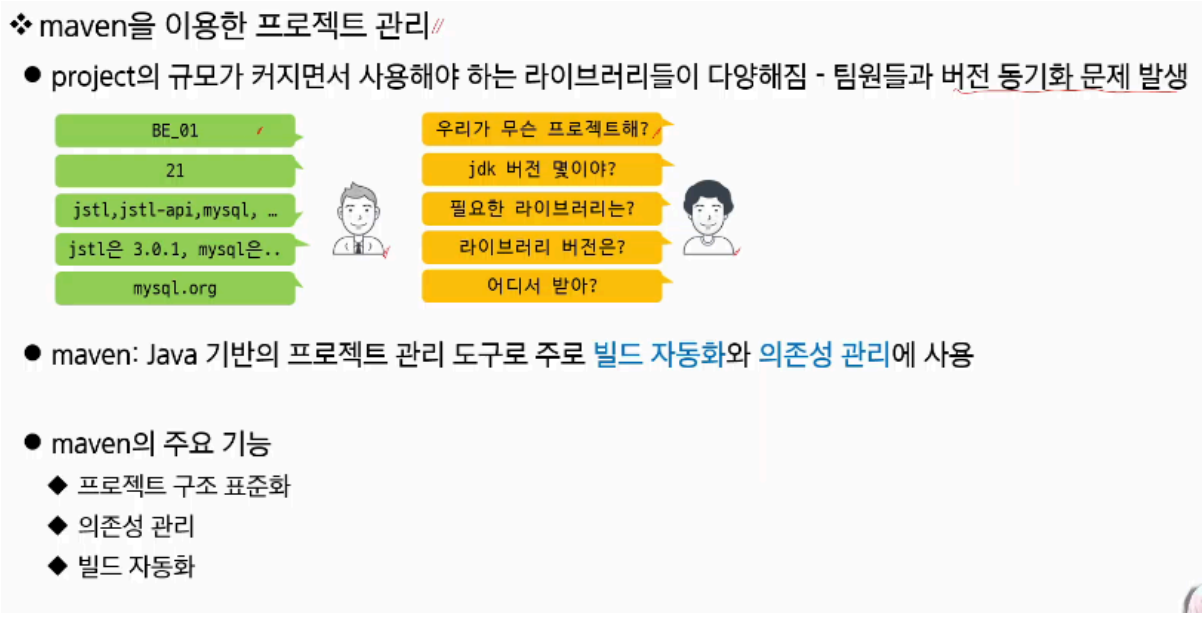
Project Facets 버전 변경

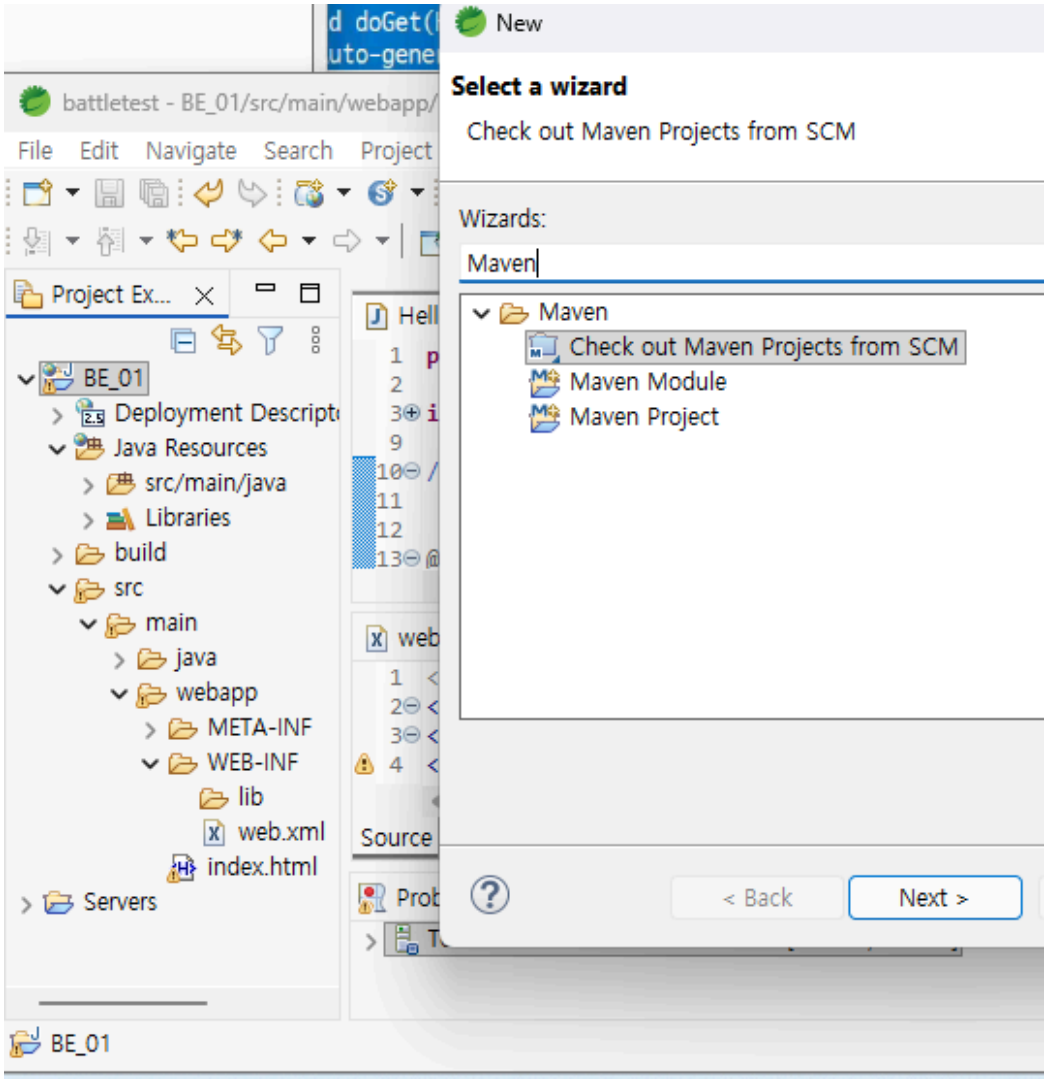
1. 동일한 Properties 창의 좌측 메뉴에서 **Project Facets**를 선택합니다.
2. 우측 목록 중 **Java** 항목의 버전을 **21**로 변경하고 우측 하단의 **Apply and Close**를 클릭합니다.

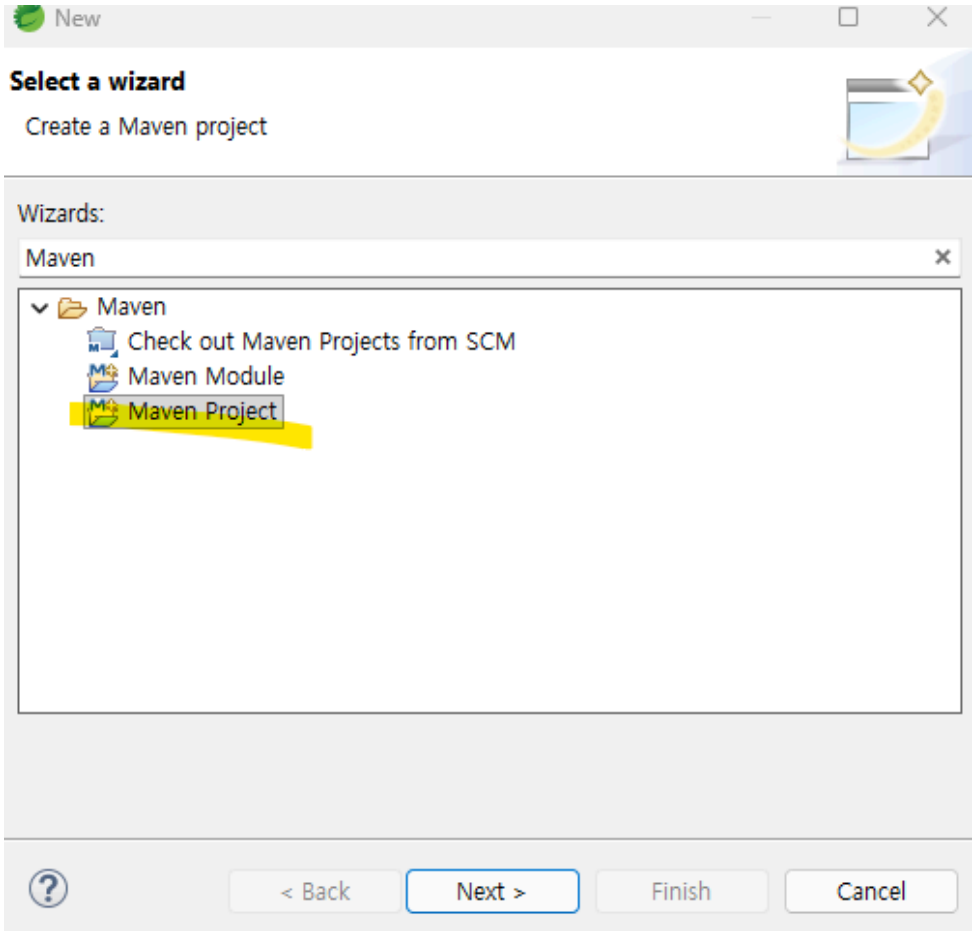
설정 완료 후 상단 메뉴에서 **Project** > Clean...을 클릭해 기존 파일을 지우고 새로 빌드한 뒤, 하단 Servers 탭에서 톰캣을 우클릭하여 **Restart**(재시작)하면 정상적으로 웹 페이지가 출력됩니다.

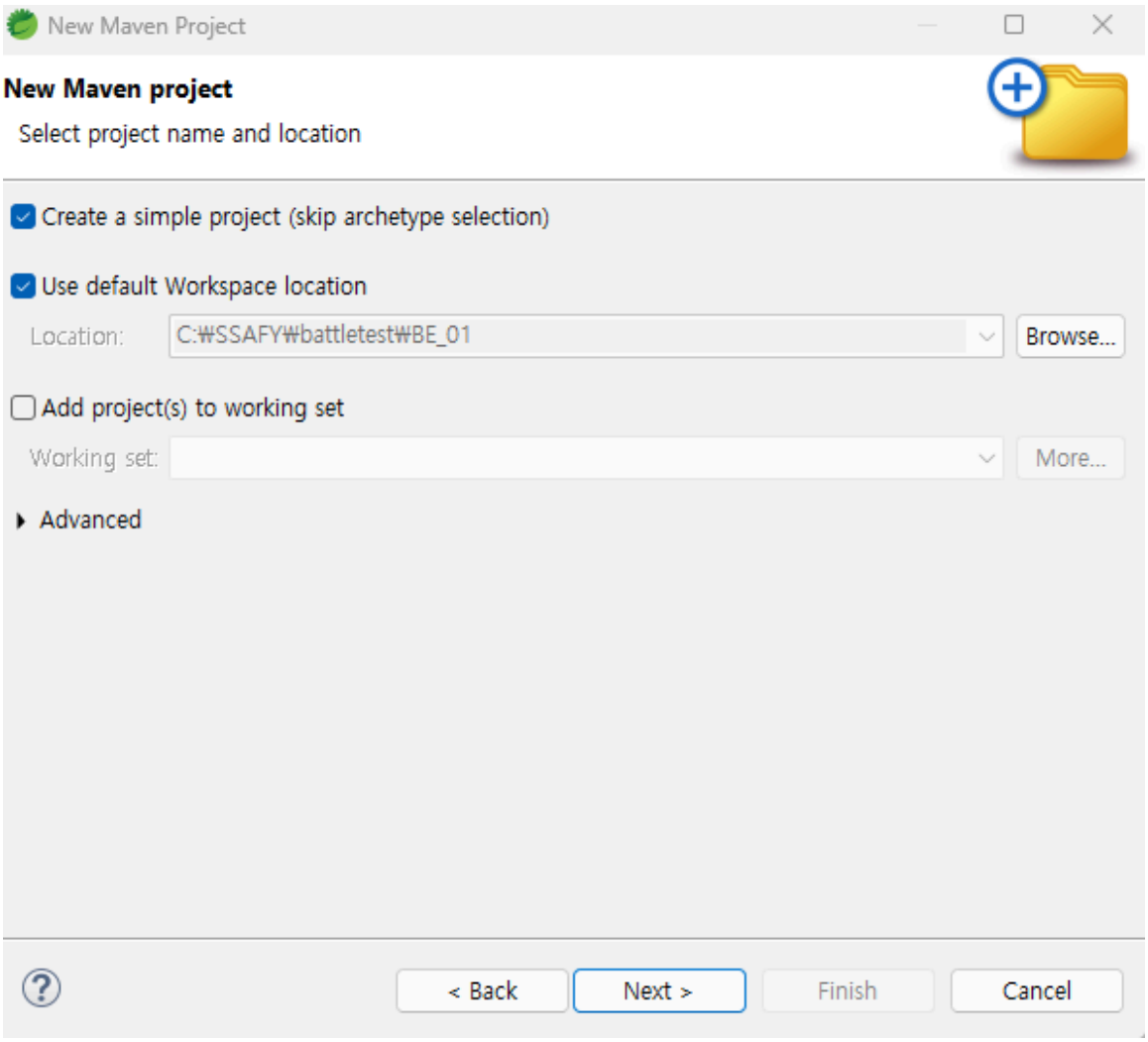


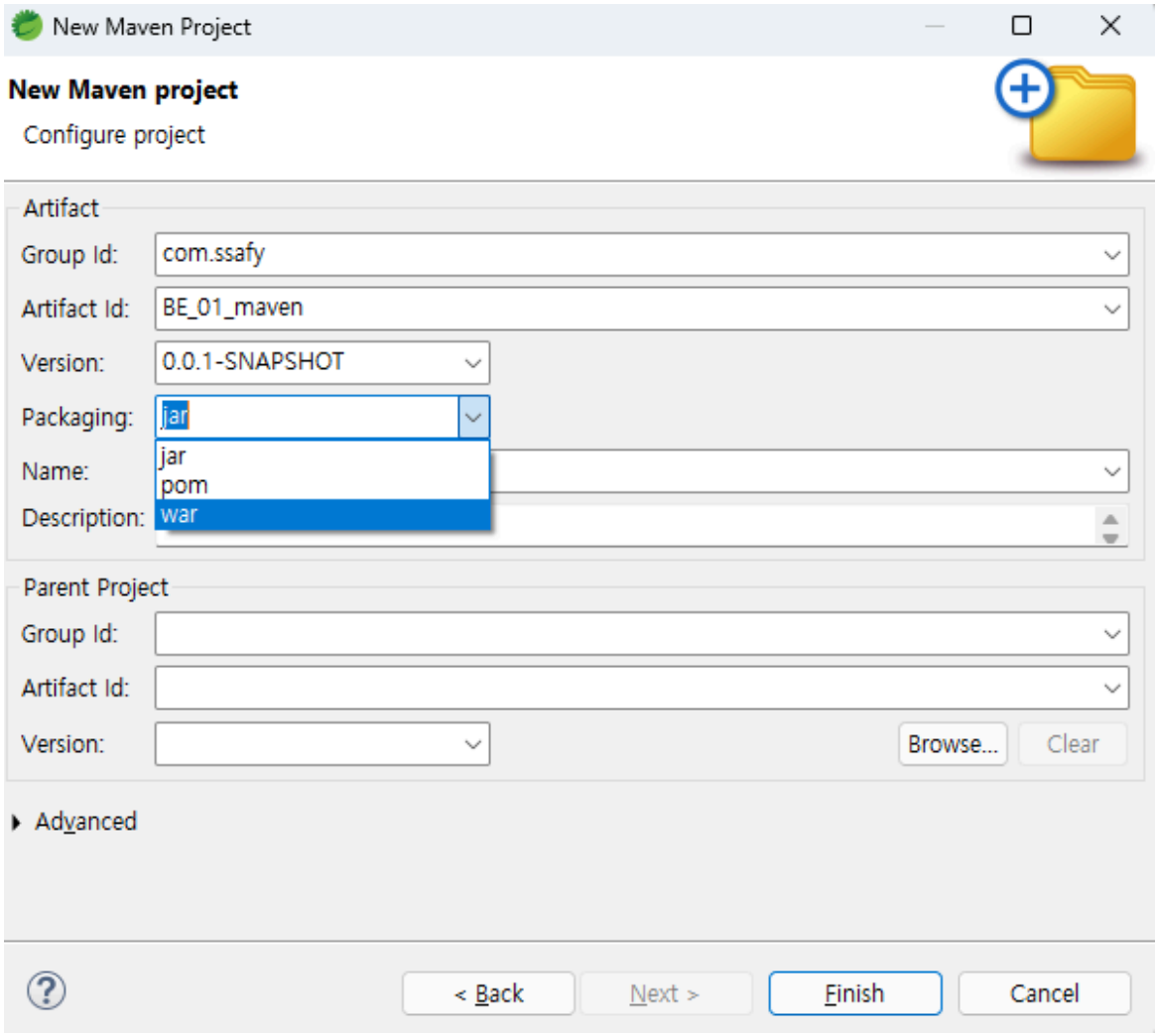
maven : project 관리하는 친구

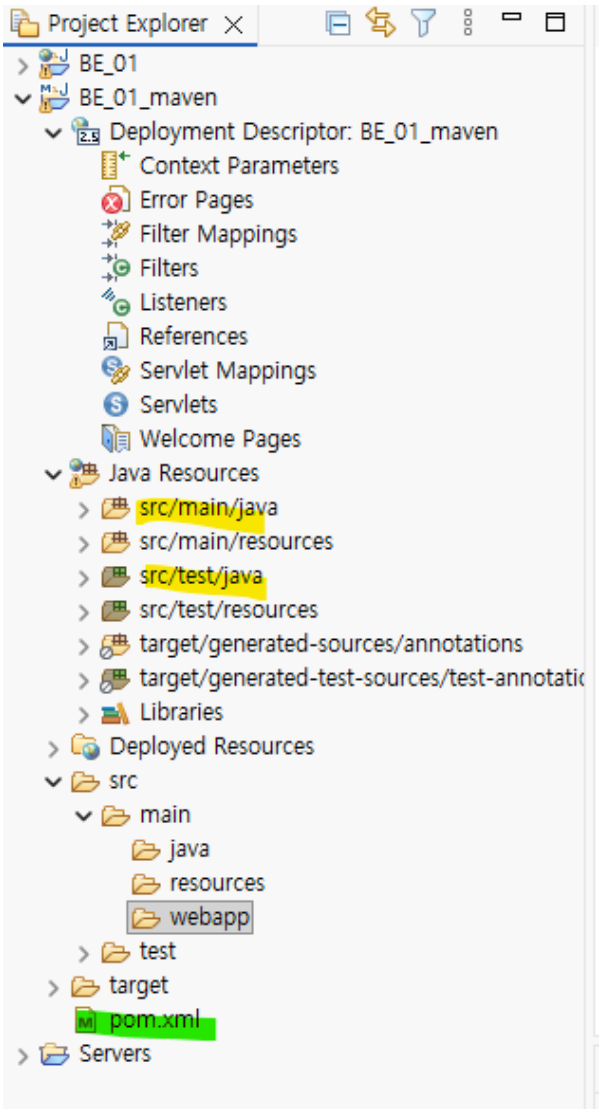



New
Select a wizard
Check out Maven Projects from SCM
Wizards:
Maven
Maven
Check out Maven Projects from SCM
Maven Module
Maven Project
< Back
Next >
battletest - BE_01/src/main/webapp/
File Edit Navigate Search Project
Project Ex...
BE_01
Deployment Descript
Java Resources
src/main/java
Libraries
build
src
main
java
webapp
META-INF
WEB-INF
lib
web.xml
index.html
Servers
Source
BE_01


New
Select a wizard
Create a Maven project
Wizards:
Maven
Maven
Check out Maven Projects from SCM
Maven Module
Maven Project
< Back
Next >
Finish
Cancel

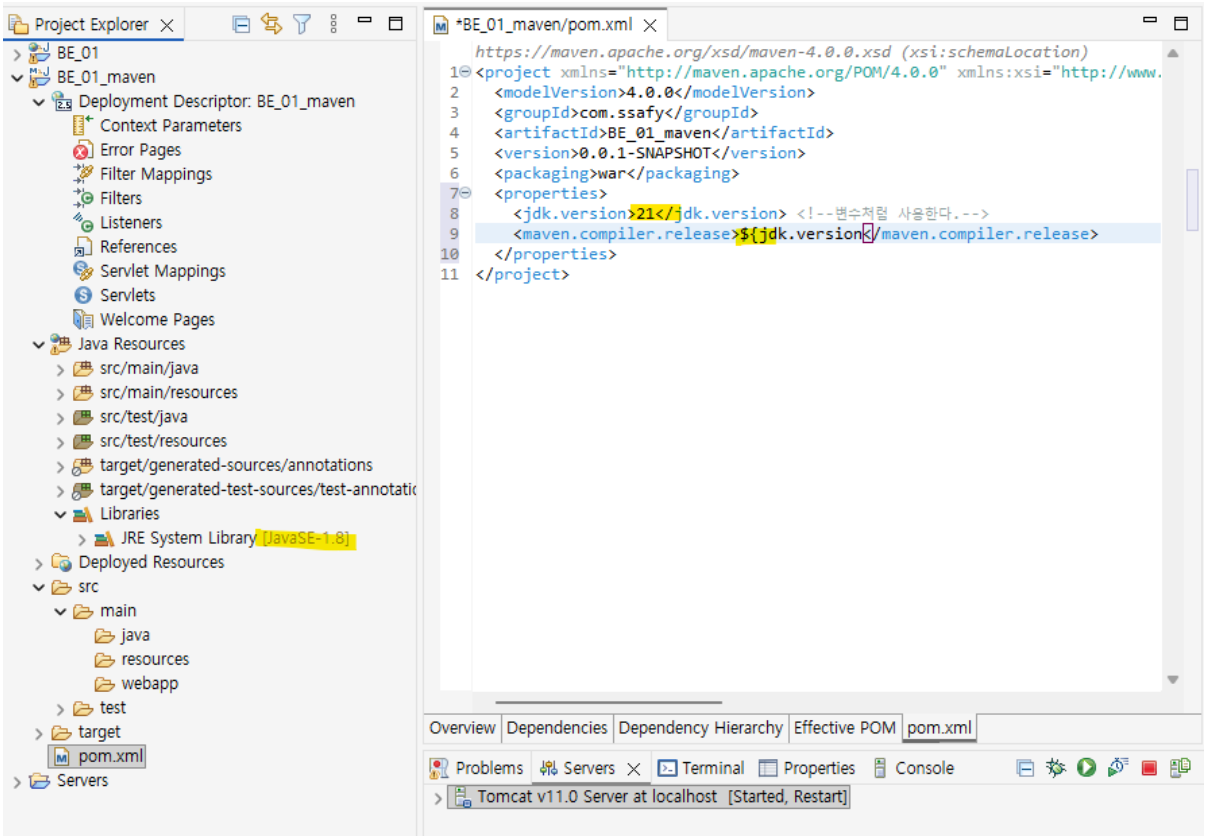

New Maven Project
New Maven project
Select project name and location
Create a simple project (skip archetype selection)
Use default Workspace location
Location:
C:₩SSAFY₩battletest₩BE_01
Browse...
Add project(s) to working set
Working set:
More...
Advanced
< Back
Next >
Finish
Cancel

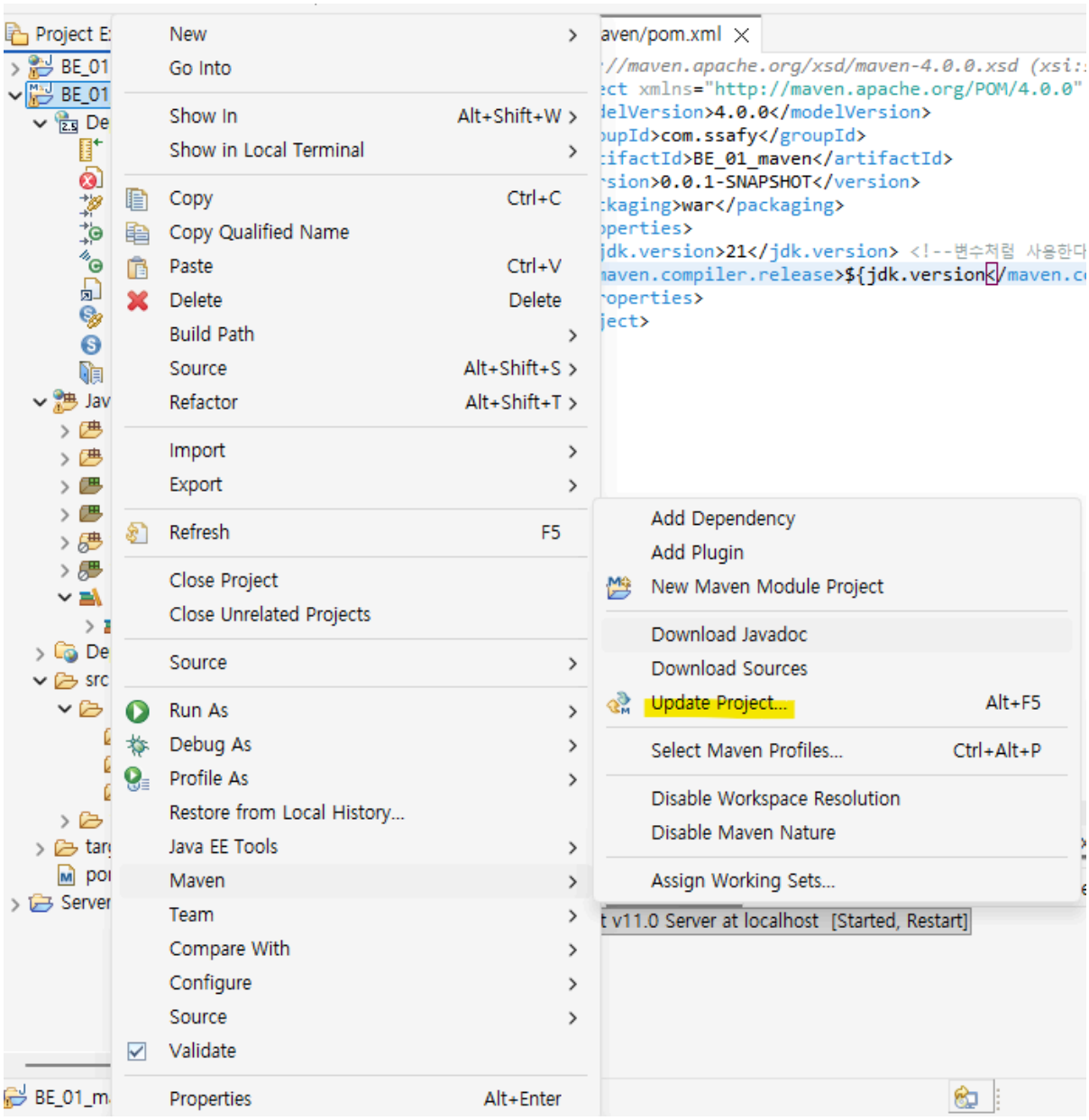

New Maven Project
New Maven project
Configure project
Artifact
Group Id:
com.ssafy
Artifact Id:
BE_01_maven
Version:
0.0.1-SNAPSHOT
Packaging:
jar
jar
pom
war
Name:
Description:
Parent Project
Group Id:
Artifact Id:
Version:
Browse...
Clear
Advanced
< Back
Next >
Finish
Cancel

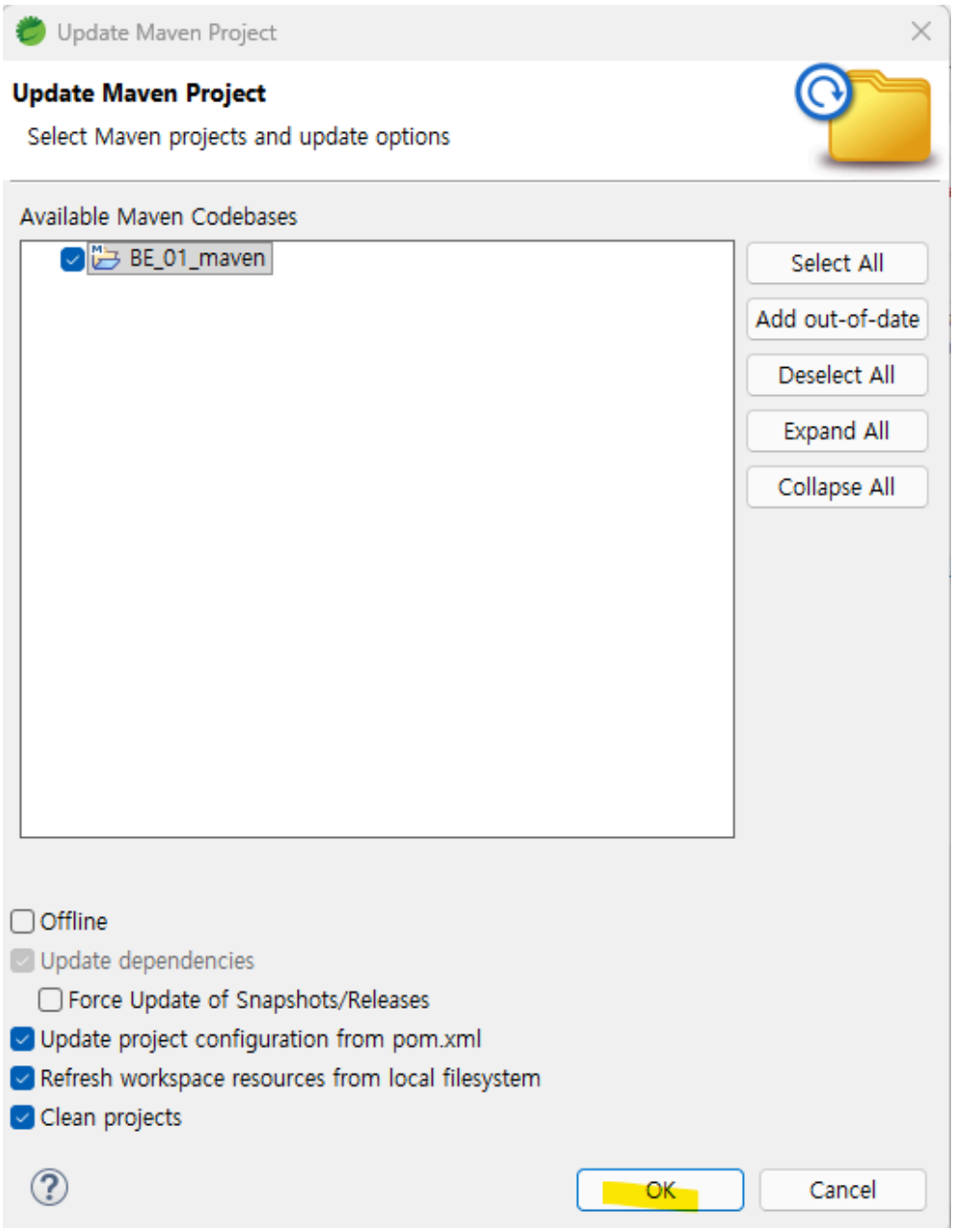


Test와 Main을 분리해놓은 이유는 무엇일까?

**분리해두는 목적이 있다.


Project Explorer
BE_01
BE_01_maven
Deployment Descriptor: BE_01_maven
Context Parameters
Error Pages
Filter Mappings
Filters
Listeners
References
Servlet Mappings
Servlets
Welcome Pages
Java Resources
src/main/java
src/main/resources
src/test/java
src/test/resources
target/generated-sources/annotations
target/generated-test-sources/test-annotati
Libraries
JRE System Library [JavaSE-1.8]
Deployed Resources
src
main
java
resources
webapp
test
target
pom.xml
Servers
*BE_01_maven/pom.xml
https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.
<modelVersion>4.0.0</modelVersion>
<groupId>com.ssafy</groupId>
<artifactId>BE_01_maven</artifactId>
<version>0.0.1-SNAPSHOT</version>
<packaging>war</packaging>
<properties>
<jdk.version>21</jdk.version> <!--변수처럼 사용한다.-->
<maven.compiler.release>${jdk.version}</maven.compiler.release>
</properties>
</project>
Overview
Dependencies
Dependency Hierarchy
Effective POM
pom.xml
Problems
Servers
Terminal
Properties
Console
Tomcat v11.0 Server at localhost [Started, Restart]


New
Go Into
Show In Alt+Shift+W
Show in Local Terminal
Copy Ctrl+C
Copy Qualified Name
Paste Ctrl+V
Delete Delete
Build Path
Source Alt+Shift+S
Refactor Alt+Shift+T
Import
Export
Refresh F5
Close Project
Close Unrelated Projects
Source
Run As
Debug As
Profile As
Restore from Local History...
Java EE Tools
Maven
Team
Compare With
Configure
Source
Validate
Properties Alt+Enter
Add Dependency
Add Plugin
New Maven Module Project
Download Javadoc
Download Sources
Update Project... Alt+F5
Select Maven Profiles... Ctrl+Alt+P
Disable Workspace Resolution
Disable Maven Nature
Assign Working Sets...
aven/pom.xml
//maven.apache.org/xsd/maven-4.0.0.xsd (xsi:
ect xmlns="http://maven.apache.org/POM/4.0.0"
delVersion>4.0.0</modelVersion>
oupId>com.ssafy</groupId>
tifactId>BE_01_maven</artifactId>
rsion>0.0.1-SNAPSHOT</version>
ckaging>war</packaging>
operties>
jdk.version>21</jdk.version> <!--변수처럼 사용한다
aven.compiler.release>${jdk.version}</maven.c
roperties>
ject>
t v11.0 Server at localhost [Started, Restart]


Update Maven Project
Update Maven Project
Select Maven projects and update options
Available Maven Codebases
BE_01_maven
Select All
Add out-of-date
Deselect All
Expand All
Collapse All
Offline
Update dependencies
Force Update of Snapshots/Releases
Update project configuration from pom.xml
Refresh workspace resources from local filesystem
Clean projects
OK
Cancel

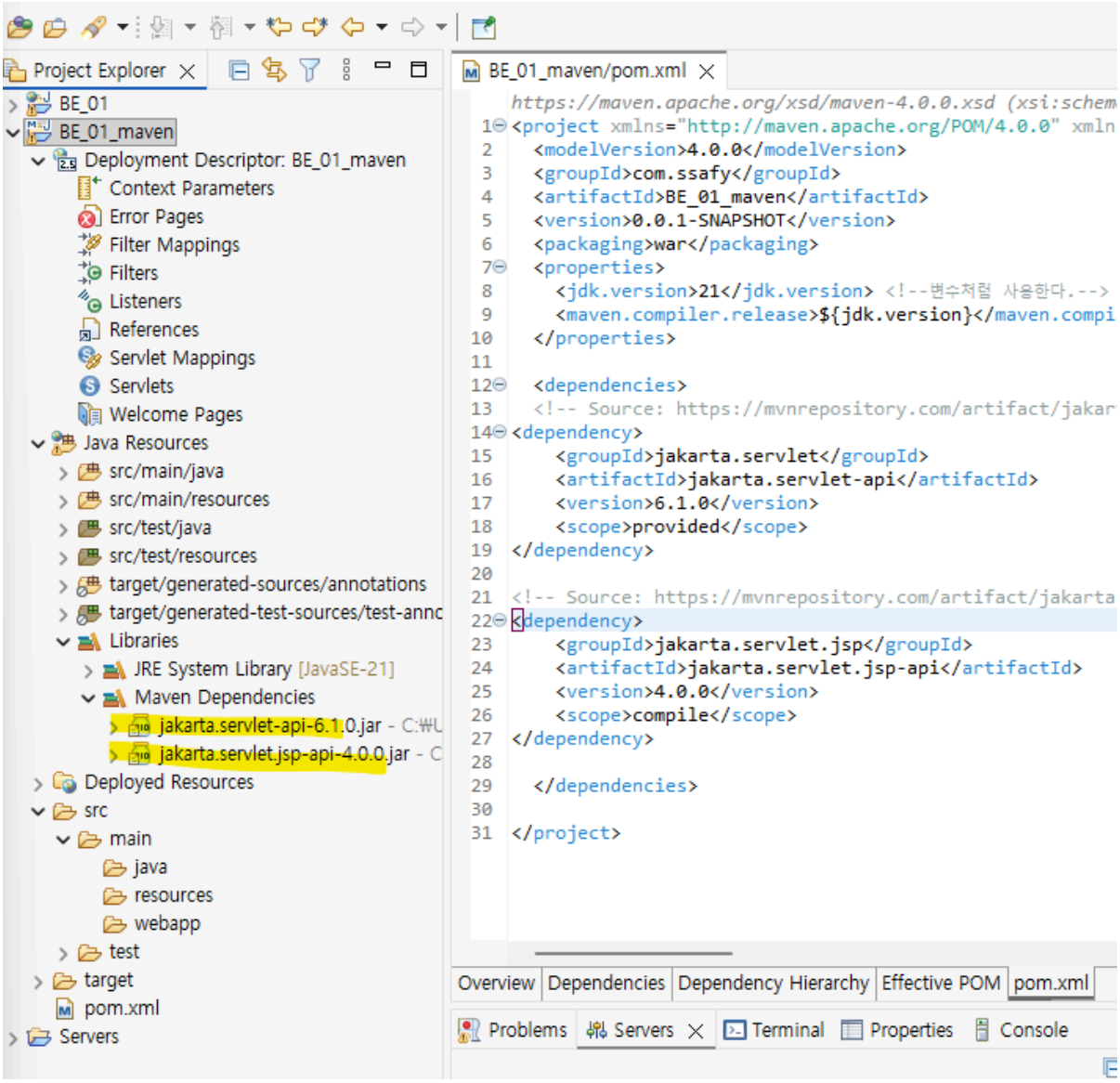

MVN REPOSITORY
Type / to search
Search
Categories
Popular
Current Group
GROUP
Jakarta Servlet
jakarta.servlet
DESCRIPTION
Jakarta Servlet
LINKS
Home » jakarta.servlet » jakarta.servlet-api
Jakarta Servlet API
API for Jakarta Servlet, enabling web applications to handle HTTP requests and responses through a standard interface for building dynamic web content.
Overview
Versions (27)
Used By (10.1K)
BOMs (1.4K)
Badges
Books (50)

6.1.0



4.0.0

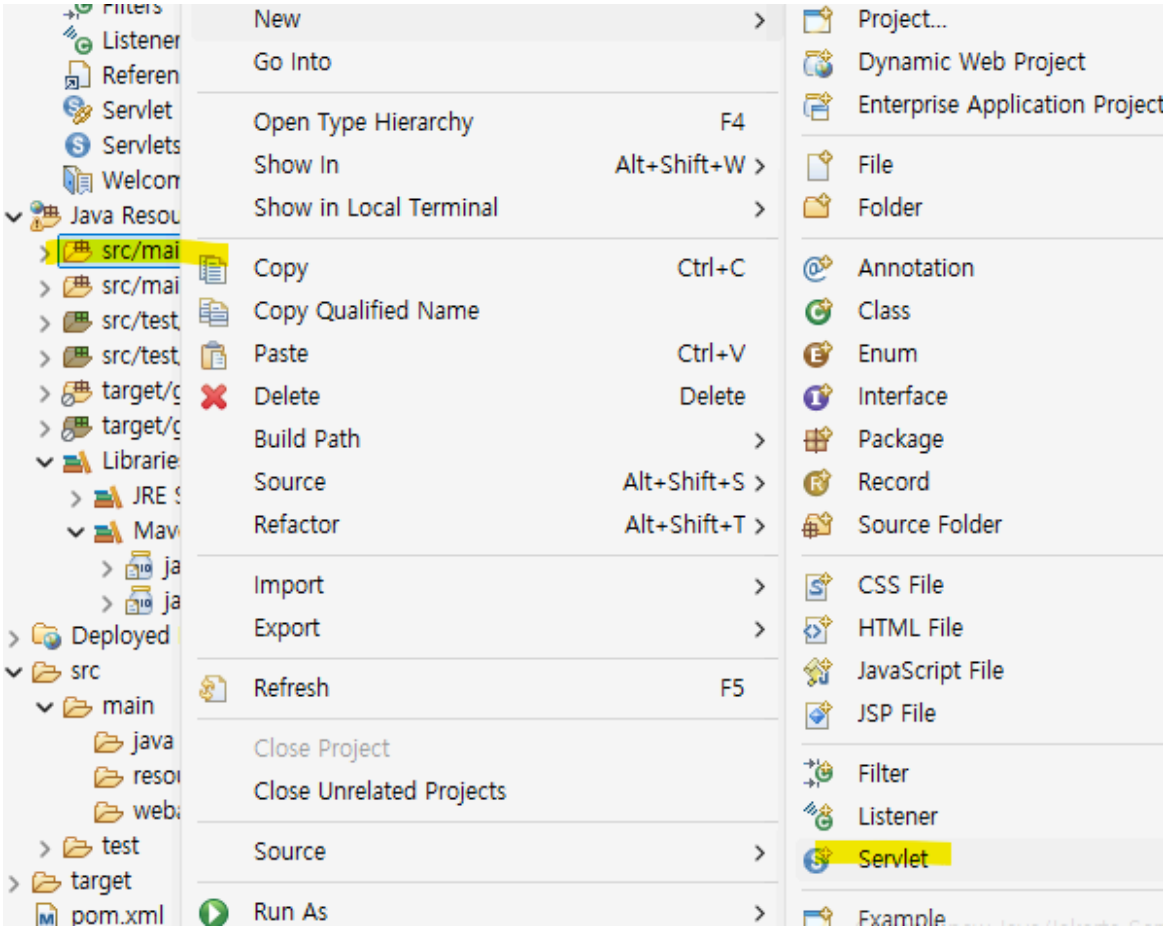
ctrl + s 후 확인



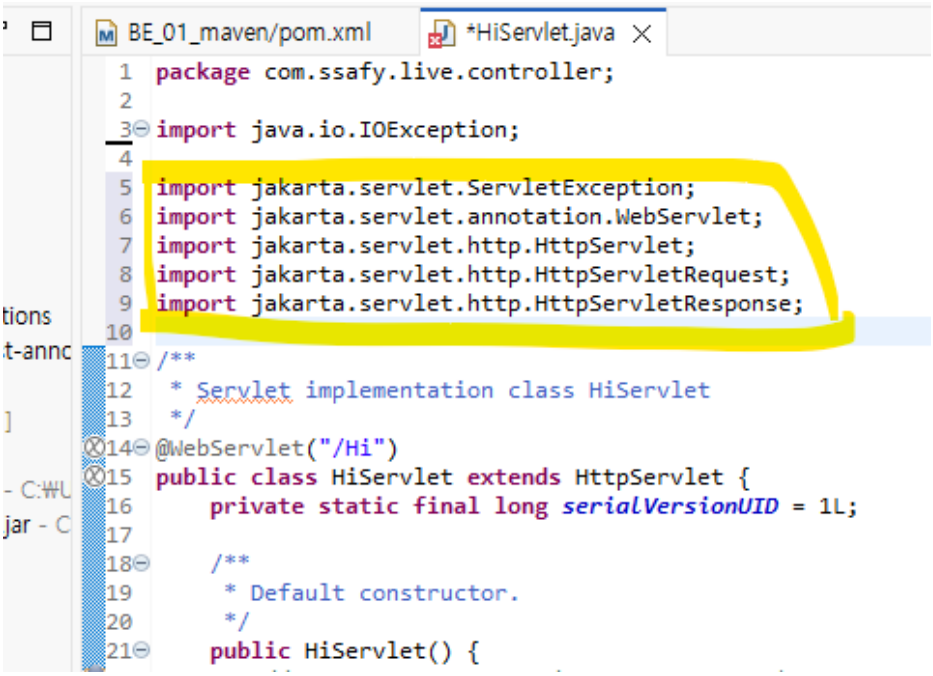
버전을 변수화해서 사용하는 것이 좋다.

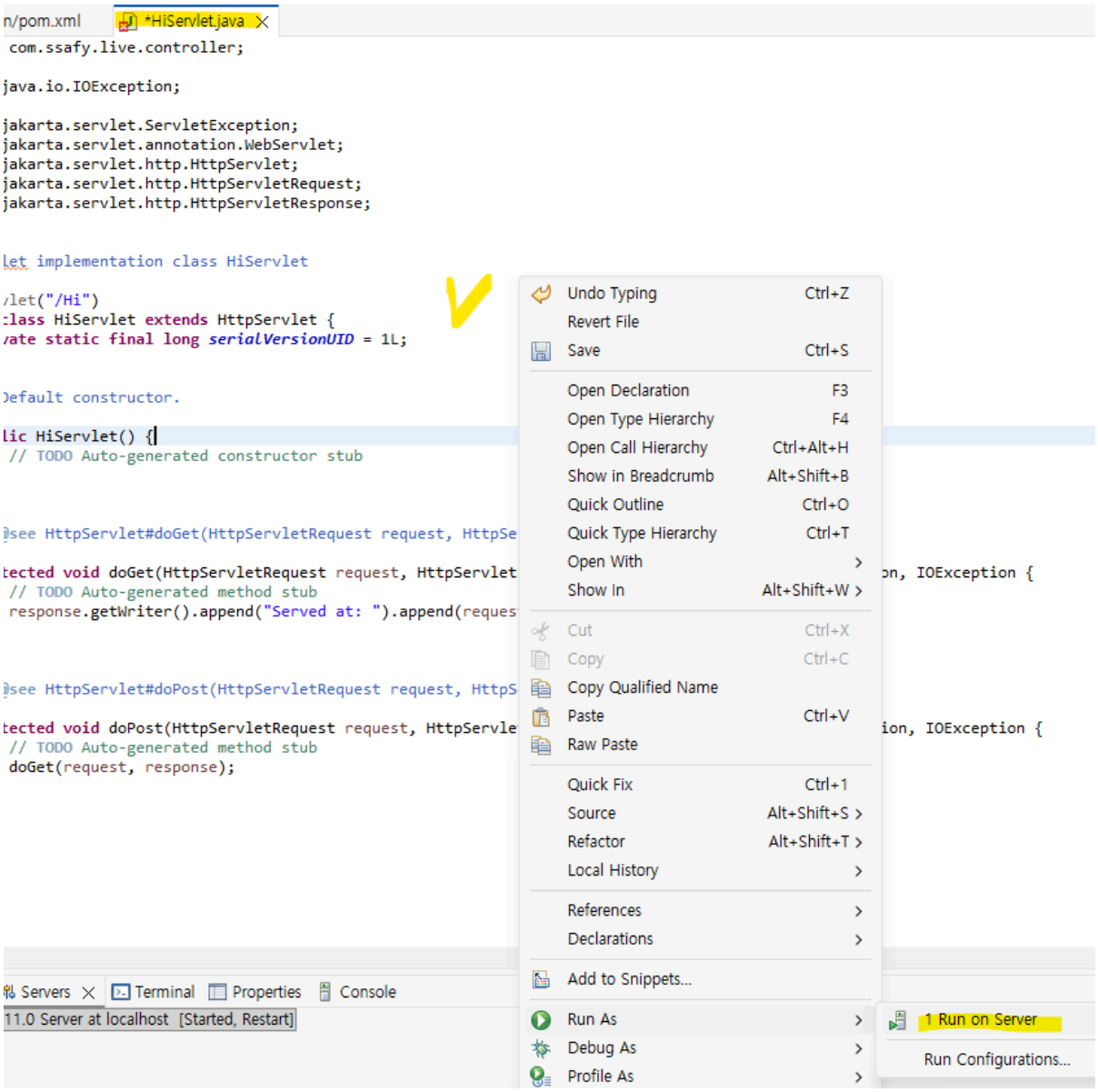
```
6    <packaging>war</packaging>
7    <properties>
8      <jdk.version>21</jdk.version> <!--변수처럼 사용한다.-->
9      <maven.compiler.release>${jdk.version}</maven.compiler.rele
10     <jakarta.servlet-api>6.1.0</jakarta.servlet-api>
11   </properties>
12
13   <dependencies>
14   <!-- Source: https://mvnrepository.com/artifact/jakarta.servl
15 <dependency>
16     <groupId>jakarta.servlet</groupId>
17     <artifactId>jakarta.servlet-api</artifactId>
18     <version>${jakarta.servlet-api}</version>
19     <scope>provided</scope>
20 </dependency>
```

main.java에 servlet 생성



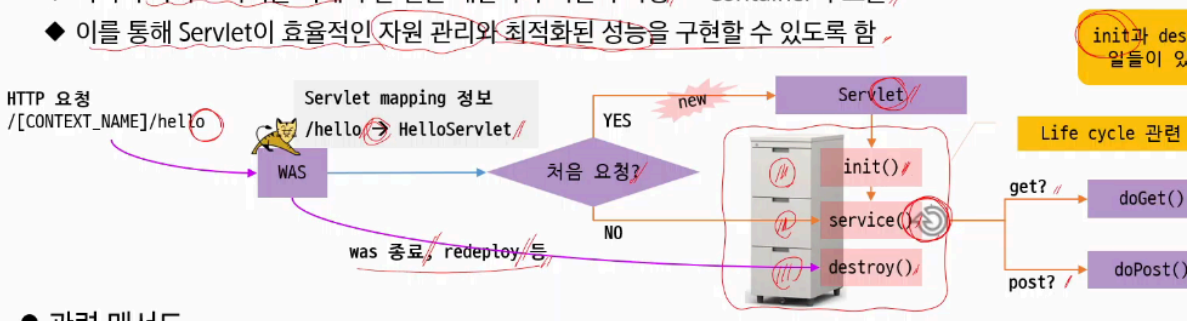
servelt 생성 후 ctrl + shift + O


BE_01_maven/pom.xml
*HiServlet.java
1 package com.ssafy.live.controller;
2
3 import java.io.IOException;
4
5 import jakarta.servlet.ServletException;
6 import jakarta.servlet.annotation.WebServlet;
7 import jakarta.servlet.http.HttpServlet;
8 import jakarta.servlet.http.HttpServletRequest;
9 import jakarta.servlet.http.HttpServletResponse;
10
11 /**
12 * Servlet implementation class HiServlet
13 */
14 @WebServlet("/Hi")
15 public class HiServlet extends HttpServlet {
16 private static final long serialVersionUID = 1L;
17
18 /**
19 * Default constructor.
20 */
21 public HiServlet() {


n/pom.xml
*HiServlet.java
com.ssafy.live.controller;
java.io.IOException;
jakarta.servlet.ServletException;
jakarta.servlet.annotation.WebServlet;
jakarta.servlet.http.HttpServlet;
jakarta.servlet.http.HttpServletRequest;
jakarta.servlet.http.HttpServletResponse;
let implementation class HiServlet
/let("/Hi")
class HiServlet extends HttpServlet {
vate static final long serialVersionUID = 1L;
Default constructor.
lic HiServlet() {
// TODO Auto-generated constructor stub
@see HttpServlet#doGet(HttpServletRequest request, HttpSe
tected void doGet(HttpServletRequest request, HttpServlet
// TODO Auto-generated method stub
response.getWriter().append("Served at: ").append(reques
on, IOException {
@see HttpServlet#doPost(HttpServletRequest request, HttpS
tected void doPost(HttpServletRequest request, HttpServle
// TODO Auto-generated method stub
doGet(request, response);
ion, IOException {
Servers
Terminal
Properties
Console
11.0 Server at localhost [Started, Restart]
Undo Typing Ctrl+Z
Revert File
Save Ctrl+S
Open Declaration F3
Open Type Hierarchy F4
Open Call Hierarchy Ctrl+Alt+H
Show in Breadcrumb Alt+Shift+B
Quick Outline Ctrl+O
Quick Type Hierarchy Ctrl+T
Open With
Show In Alt+Shift+W
Cut Ctrl+X
Copy Ctrl+C
Copy Qualified Name
Paste Ctrl+V
Raw Paste
Quick Fix Ctrl+1
Source Alt+Shift+S
Refactor Alt+Shift+T
Local History
References
Declarations
Add to Snippets...
Run As
Debug As
Profile As
1 Run on Server
Run Configurations...

❖ Servlet Life Cycle 관리

- 개발자는 Servlet을 만들지만 객체를 만들거나 호출하지 않음 - Container가 life cycle에 따라 관리
  - ◆ 각각의 라이프 사이클 훅에서 할 일을 개발자가 적절히 작성 → Container가 호출
  - ◆ 이를 통해 Servlet이 효율적인 자원 관리와 최적화된 성능을 구현할 수 있도록 함



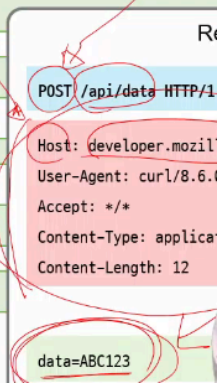
- 관련 메서드

메서드명	설명	용도
init()	어떤 요청도 init이 종료되기 전에는 처리될 수 없음	Servlet에서 필요한 자원 초기화
service()	실제 사용자의 요청을 처리하는 메서드	요청 방식에 따라 doGet(), doPost()
destroy()	어떤 요청이라도 처리하고 있으면 destroy는 동작하지 않음	init에서 초기화한 자원의 정리 작업

❖ HttpServletRequest

- HTTP 요청을 추상화한 인터페이스로 헤더, 파라미터, 속성 및 요청 본문과 관련된 정보에 접근하는

메서드	설명
String getMethod()	요청이 만들어진 HTTP 메서드의 이름을 반환
String getParameter(String name)	요청 파라미터의 값 반환
String[] getParameterValues(String name)	지정된 요청 파라미터의 모든 값을 포함하는 배열 반환
Enumeration<String> getParameterNames()	요청에 포함된 파라미터 이름의 열거형 반환
String getHeader(String name)	지정된 요청 헤더의 값을 반환
Enumeration<String> getHeaderNames()	요청에 포함된 모든 헤더 이름의 열거형 반환
void setAttribute(String name, Object o)	요청에 속성을 저장
Object getAttribute(String name)	요청에 저장된 속성의 값을 반환
HttpSession getSession()	현재 요청과 연관된 세션 반환



멱등 : 특정 파라미터를 여러번 계산해도 값이 동일하다.

❖ HTTP method: form의 데이터를 서버로 전달하는 방식 결정

HTTP 메소드	RFC	요청에 Body가 있음	응답에 Body가 있음	안전	멱등(Idempotent)	캐시 가능
GET	RFC 9110	선택 사항	예	예	예	예
POST	RFC 9110	예	예	아니요	아니요	예
PUT	RFC 9110	예	예	아니요	예	아니요
DELETE	RFC 9110	선택 사항	예	아니요	예	아니요
CONNECT	RFC 9110	선택 사항	예	아니요	아니요	아니요
OPTIONS	RFC 9110	선택 사항	예	예	예	아니요
TRACE	RFC 9110	아니요	예	예	예	아니요
PATCH	RFC 5789	예	예	아니요	아니요	아니요
HEAD	RFC 9110	선택 사항	아니요	예	예	예

❖ Request parameter

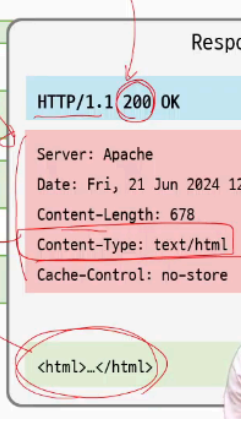
- <form> 또는 queryString을 통해서 클라이언트가 전달한 값으로 언제나 문자열
  - ◆ 처음 클라이언트에서 설정된 이후 조작은 불가

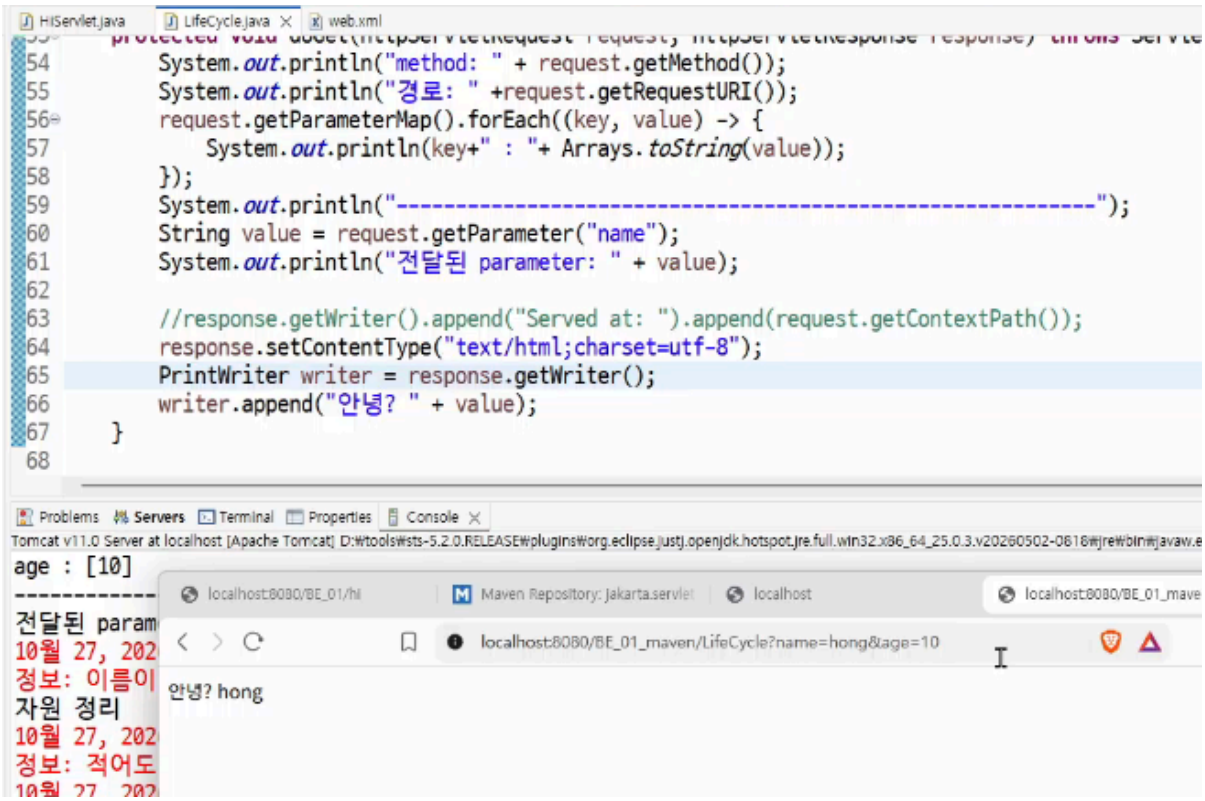


❖ HttpServletResponse

- HTTP 응답을 추상화한 인터페이스로 헤더 조회/설정, 응답 상태, 응답 전송을 위한 작업 처리

메서드명	반환타입	설명
setStatus()	void	응답 상태 코드 설정
setHeader()	void	HTTP 응답 헤더 설정
setContentType()	void	응답 콘텐츠 타입과 함께 encoding도 설정
getWriter()	PrintWriter	문자 응답 스트림
addCookie()	void	HTTP 응답에 쿠키 추가
sendRedirect()	void	지정된 URL로 리다이렉트
sendError()	void	에러 응답 전송





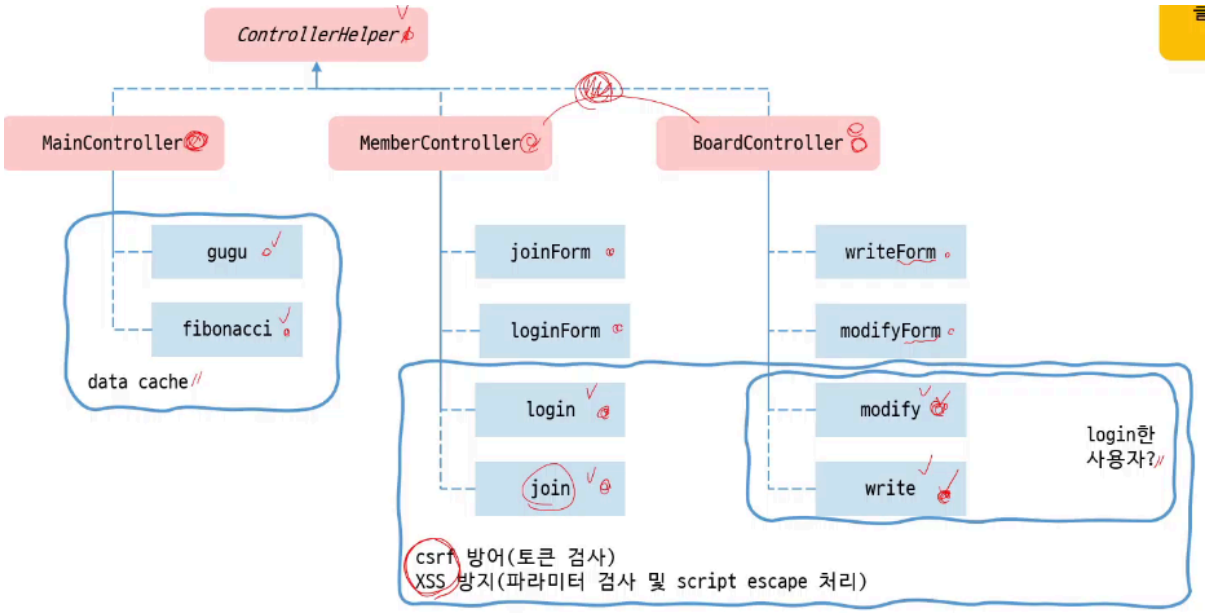
AddServlet.java

```
10
11 *
12  Servlet implementation class AddServlet
13 /
14 ebServlet("/add")
15 blic class AddServlet extends HttpServlet {
16   private static final long serialVersionUID = 1L;
17
18
19   /**
20    * @see HttpServlet#doGet(HttpServletRequest request, HttpServletResponse respo
21    */
22   protected void doGet(HttpServletRequest request, HttpServletResponse response)
23       String num1 = request.getParameter("num1");
24       String num2 = request.getParameter("num2");
25
26       int result = Integer.parseInt(num1) + Integer.parseInt(num2);
27
28       response.setContentType("text/html;charset=utf-8");
29       response.getWriter()
30       .append("<html><body> 더하기 결과는 " + result+"</body></html>");
31   }
32
```

HttpServlet, HttpServletRequest, HttpServletResponse

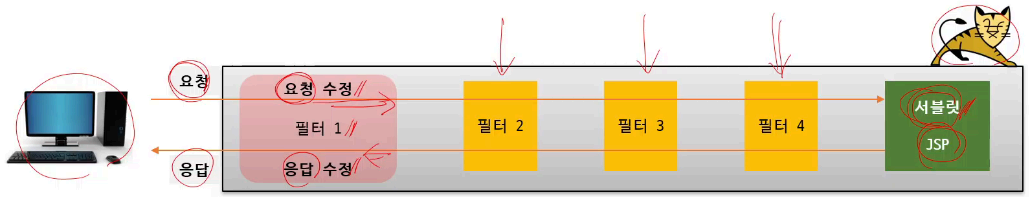
▼ 10/27 오후

1. Front Controller Pattern
2. Filter
3. Listener
4. Servlet과 Multi-Thread



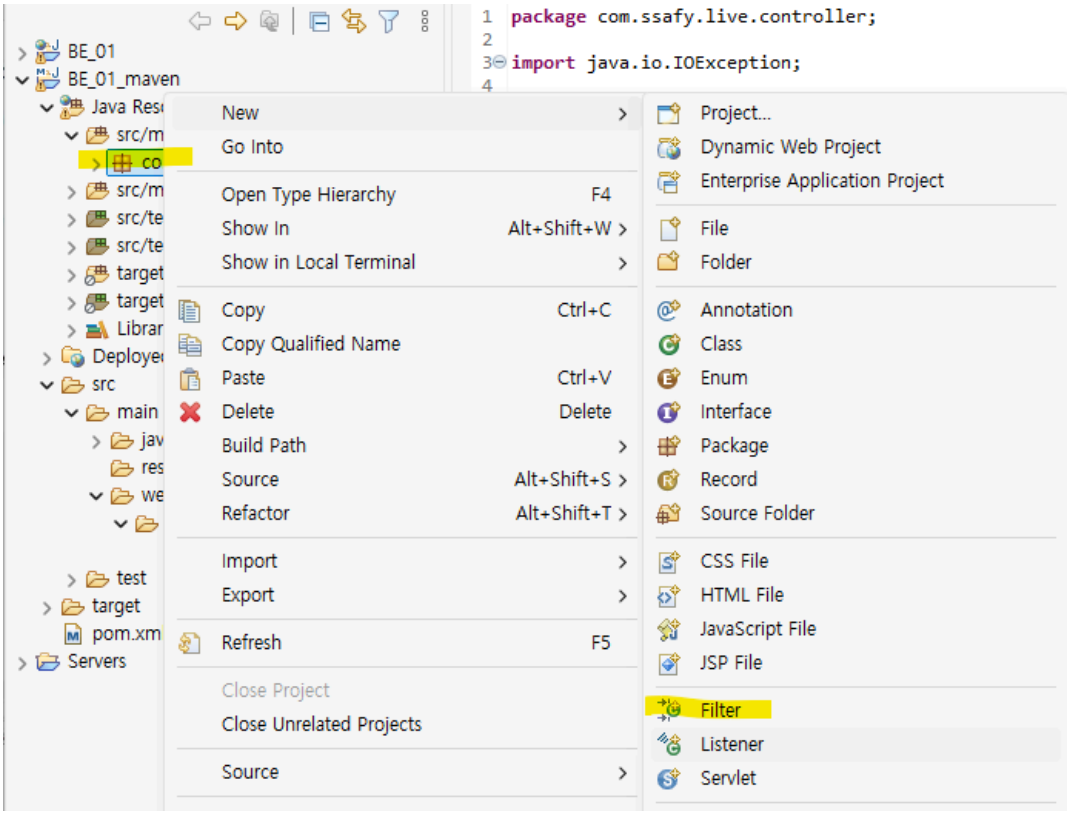
❖ Filter와 Filter Chain

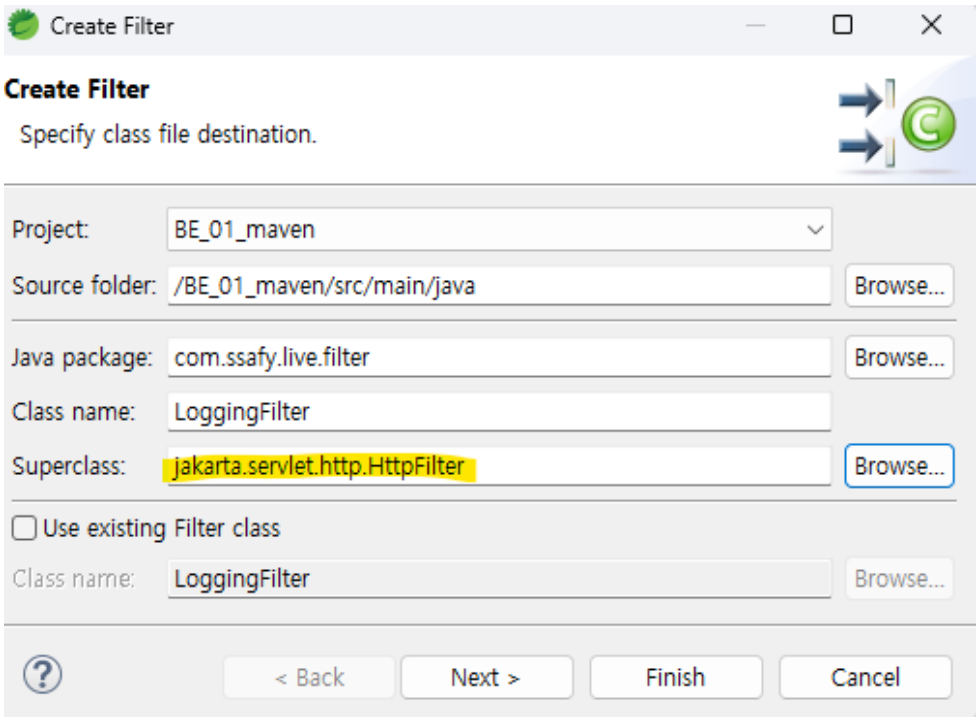
- Filter : 웹 컴포넌트가 실행되기 전/후에 요청과 응답을 가로채서 부가적인 기능을 수행하기 위한 웹 컴포넌트

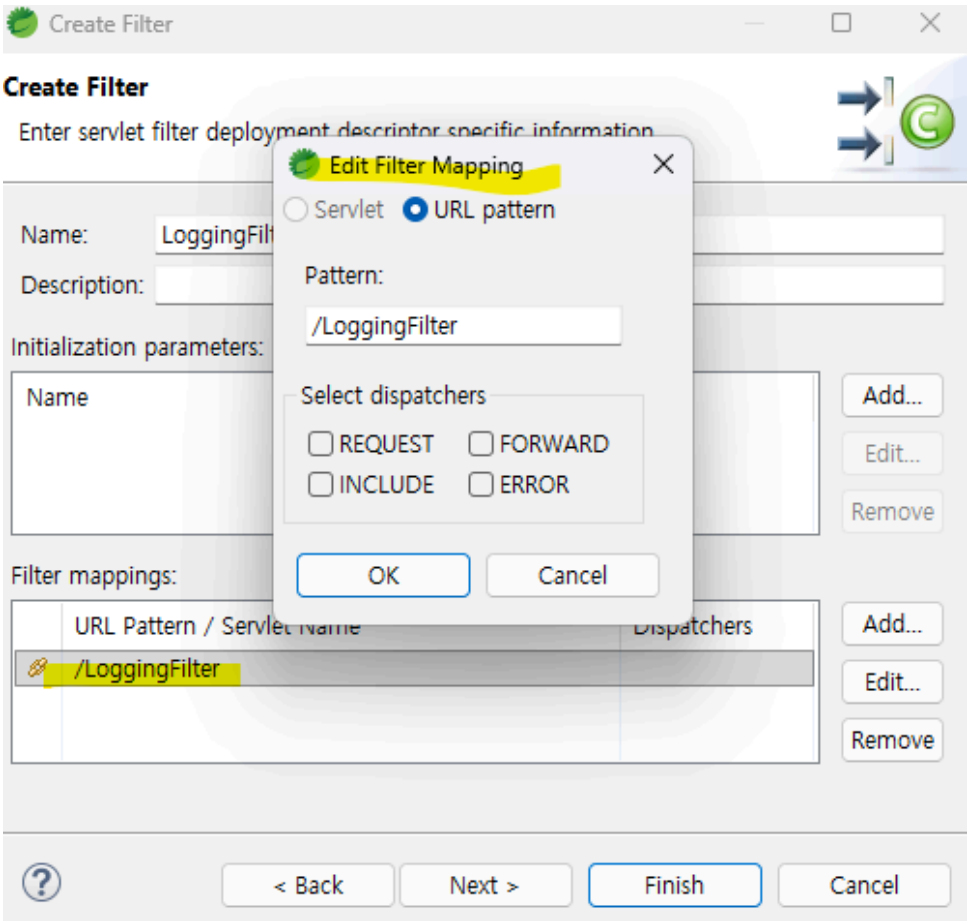


- Filter Chain : 여러 개의 필터가 모여 필터 체인 구성 가능

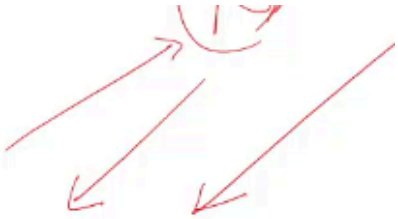
필터 만들기


1 package com.ssafy.live.controller;
2
3 import java.io.IOException;
4
BE_01
BE_01_maven
New
Go Into
Open Type Hierarchy F4
Show In Alt+Shift+W
Show in Local Terminal
Copy Ctrl+C
Copy Qualified Name
Paste Ctrl+V
Delete Delete
Build Path
Source Alt+Shift+S
Refactor Alt+Shift+T
Import
Export
Refresh F5
Close Project
Close Unrelated Projects
Source
Project...
Dynamic Web Project
Enterprise Application Project
File
Folder
Annotation
Class
Enum
Interface
Package
Record
Source Folder
CSS File
HTML File
JavaScript File
JSP File
Filter
Listener
Servlet


Create Filter
Create Filter
Specify class file destination.
Project: BE_01_maven
Source folder: /BE_01_maven/src/main/java
Browse...
Java package: com.ssafy.live.filter
Browse...
Class name: LoggingFilter
Superclass: jakarta.servlet.http.HttpFilter
Browse...
Use existing Filter class
Class name: LoggingFilter
Browse...
< Back
Next >
Finish
Cancel


Create Filter
Create Filter
Enter servlet filter deployment descriptor specific information
Edit Filter Mapping
Servlet
URL pattern
Pattern:
/LoggingFilter
Select dispatchers
REQUEST
FORWARD
INCLUDE
ERROR
OK
Cancel
Name:
Description:
Initialization parameters:
Name
Add...
Edit...
Remove
Filter mappings:
URL Pattern / Servlet Name
Dispatchers
/LoggingFilter
Add...
Edit...
Remove
< Back
Next >
Finish
Cancel

❖ Filter Mapping



- Filter 적용 대상 설정
- Servlet: filter를 적용할 개별 servlet 등록
- URL Pattern: url 기반으로 filter 적용 대상 설정
  - ◆ servlet의 url pattern과 동일
  - ◆ 경로 기반, 확장자 기반
- select dispatcher: filter를 적용할 시점
  - ◆ request : 클라이언트 직접 요청인 경우 적용(기본 값)
    - forward()는 동작 하지 않음
  - ◆ forward: forward()를 통해 이동하는 경우에 적용
  - ◆ include: include()를 통해 포함하는 경우에 적용
  - ◆ error: 에러 페이지로 전달 시
    - web.xml에 error-page 설정 필요

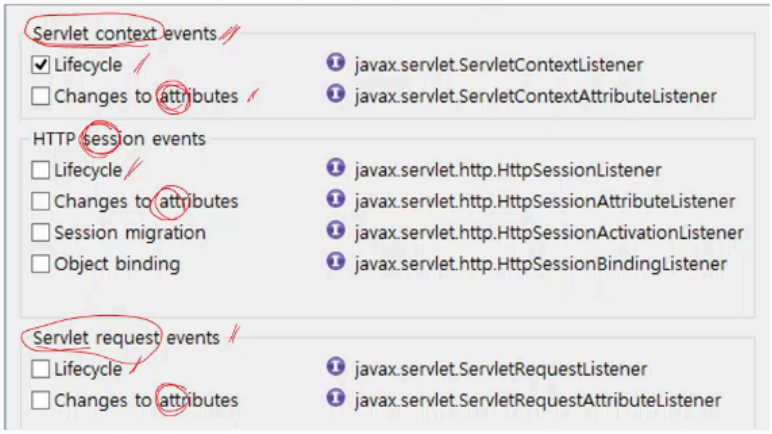
```java
/**
 * @see Filter#doFilter(ServletRequest, ServletResponse, FilterChain)
 */
public void doFilter(HttpServletRequest request, HttpServletResponse response, FilterChain chain) throws IOException, ServletException {
    // TODO Auto-generated method stub
    // place your code here

    // pass the request along the filter chain
    chain.doFilter(request, response);
}
```

프로그램은 다 돌고 도는 것

❖ Listener

- 웹 애플리케이션에서 발생하는 이벤트에 대한 모니터링 객체
- 웹 애플리케이션에서 발생하는 이벤트?



- ServletContextListener: 웹 애플리케이션 생성에서 소멸까지의 주요 사항 모니터링
  - 개별 서블릿 동작 전에 초기화하는데 비용이 많이 드는 공유 자원의 초기화에 주로 사용

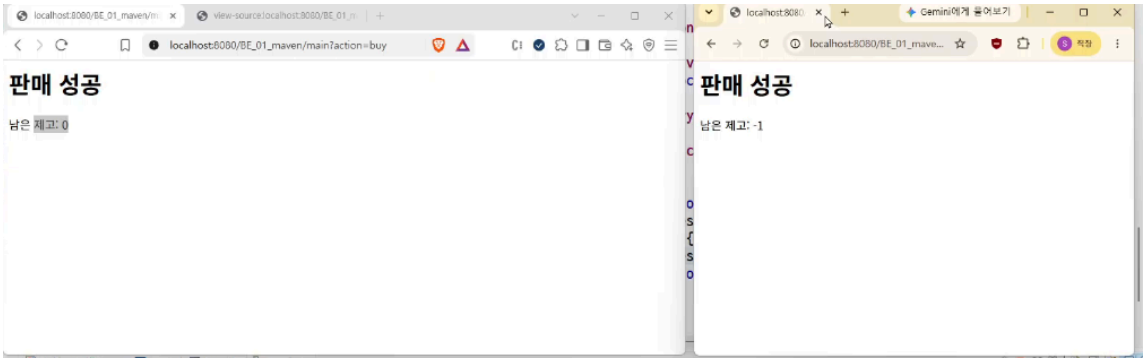
ControllerHelper.java | LifeCycle.java | index.html | XSSFilter.java | MyContextListener.java

```
 9  *
10  */
11 @WebListener
12 public class MyContextListener implements ServletContextListener {
13
14     /**
15      * @see ServletContextListener#contextInitialized(ServletContextEvent)
16      */
17     public void contextInitialized(ServletContextEvent sce)  {
18         System.out.println("컨텍스트 초기화중 - 모든 서블릿에서 사용하는 자원 초기화");
19     }
20
21     /**
22      * @see ServletContextListener#contextDestroyed(ServletContextEvent)
23      */
24     public void contextDestroyed(ServletContextEvent sce)  {
25         System.out.println("컨텍스트 정리중 - init과정에서 초기화한 리소스 정리");
26     }
27
28 }
```

❖ Servlet의 특징 돌아보기

- Servlet의 장점
  - ◆ java의 OOP 기반으로 작성:
    - 유지보수성 및 재활용성 우수하며 플랫폼 독립적
  - ◆ 높은 성능:
    - 하나의 서블릿 객체만 생성하며 멀티스레딩을 지원하여 동시 요청 처리가 가능
  - ◆ 확장성:
    - 필터를 통한 공통 모듈의 전/후 처리, 리스너를 이용한 이벤트 기반 처리 가능 및 스프링 등 다양한 프레임워크와 통합 용이

- Servlet 단점
  - ◆ business logic과 presentation logic(HTML code)가 섞여서 나타남



▼ 10/27 기타

관련 import 구문 자동 생성 / Ctrl + Shift + O

**월말평가 볼 때 서버 만들어야한다.

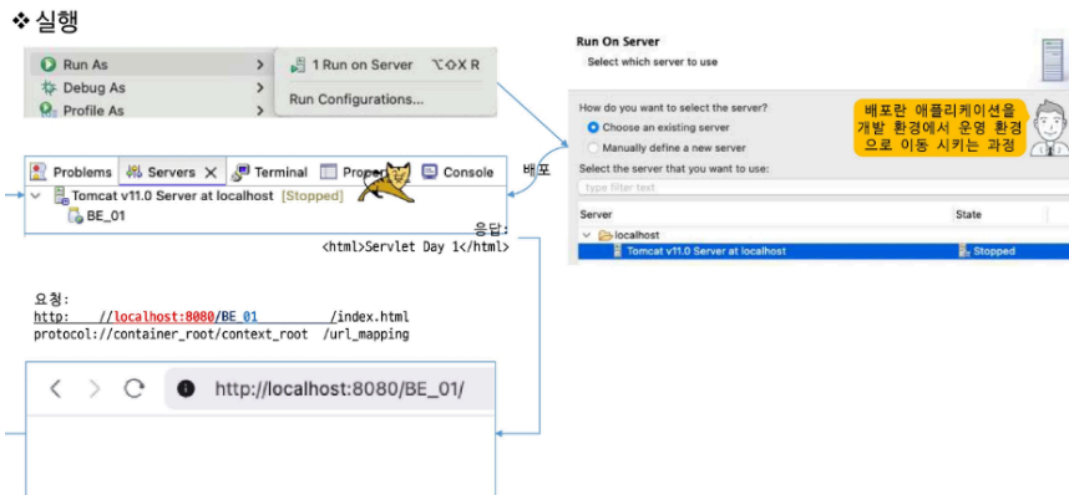
❖ Deployment descriptor

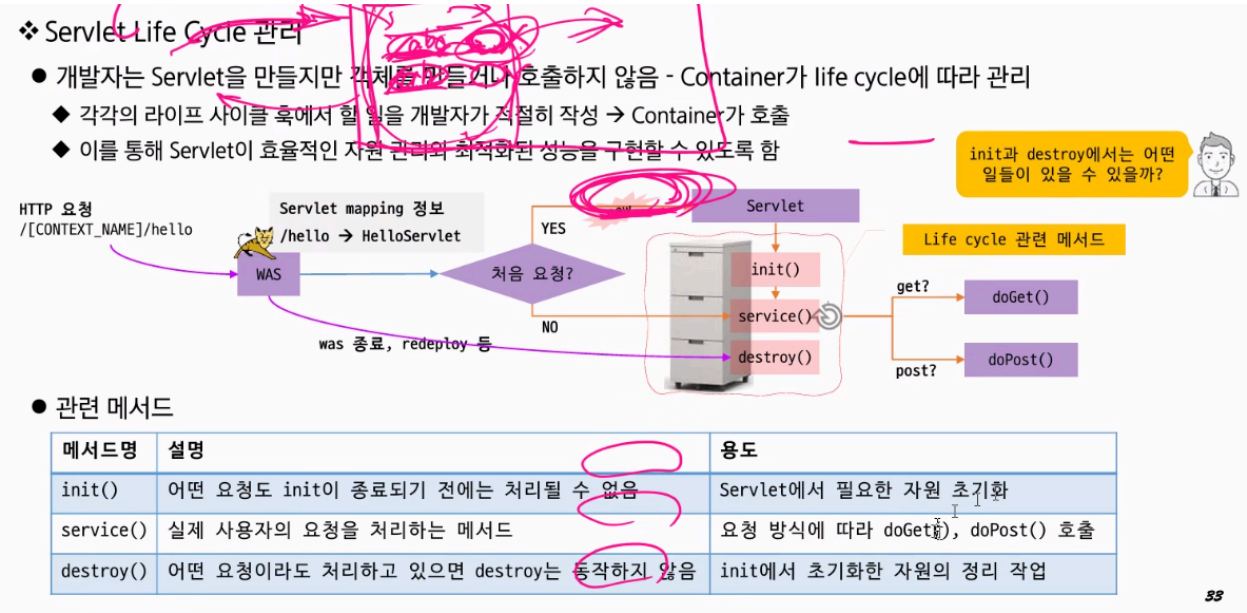
- web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="https://jakarta.ee/xml/ns/jakartaee"
xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
id="WebApp_ID" version="6.0">
<display-name>HelloServlet</display-name>
<welcome-file-list>
  <welcome-file>index.html</welcome-file>
  <welcome-file>index.jsp</welcome-file>
  <welcome-file>index.htm</welcome-file>
  <welcome-file>default.html</welcome-file>
  <welcome-file>default.jsp</welcome-file>
  <welcome-file>default.htm</welcome-file>
</welcome-file-list>
</web-app>
```

별도로 path를 입력하지 않고 context를 방문했을 때 보여지는 파일

index.html을 어디에 만든다고??





순서에 대한 문제, get 방식, post 방식 (빈출)



❖ Http Status

● 웹 서버가 클라이언트의 요청을 처리한 결과를 카테고리 별로 지정된 숫자 코드로 반환

Category	Status Code	Name	Description
2xx Success	200	OK	요청의 정상 처리 상태
	201	Created	새로운 리소스 생성의 성공 상태
3xx Redirection	302	Moved Temporarily (Found)	요청한 리소스의 임시적 위치 이동 상태 (요청한 리소스가 다른 경로에서 발견 되었다. - found elsewhere)
4xx Client Error	400	Bad Request	잘못된 문법으로 인한 서버의 요청 처리 불가 상태
	401	Unauthorized	인증 실패 (당신이 누구인지 모르겠어!)
	403	Forbidden	권한 없음 (당신이 누구인지는 알지만 이건 못해!)
	404	Not Found	요청한 리소스를 찾을 수 없는 상태
	405	Method Not Allowed	허용되지 않은 HTTP 메서드 사용 상태
5xx Server Error	500	Internal Server Error	서버 내부 오류 발생 상태

코드의 의미, 예시, 설명하시오. 서술형(빈출)